

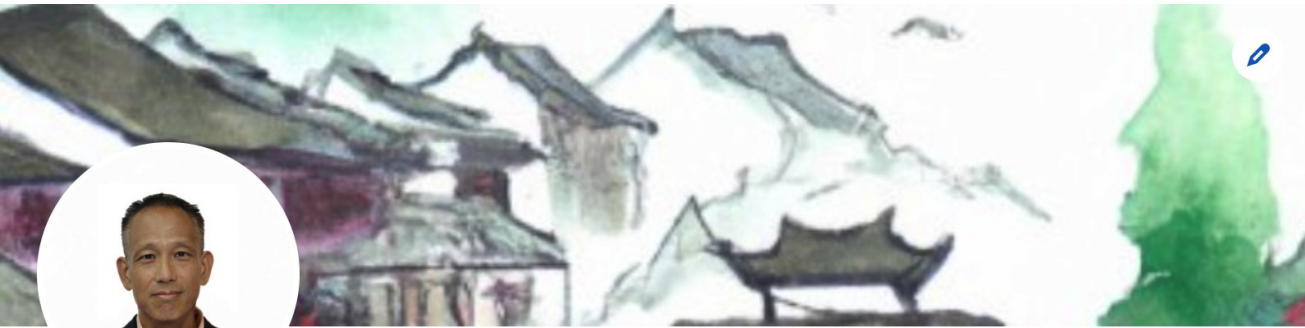
PACKET EVIDENCE • PRACTICAL DIAGNOSTICS

Wireshark Network Analysis Masterclass

C1123 | 4 days | 30 instructional hours
v4.0 • 2 October 2026
Trainer: Dr. Alfred Ang



About the Trainer



Dr. Alfred Ang ✓

DACE, ACTA, ACIT, PMP®, CSM®, GCWPL, CAIEG, Unity Certified User, Neo4J Certified Professional, AWS CLF-C02, Microsoft Certified Power BI , Power App and Automate Developer, Azure Data Scientist, AI Engineer.

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How You'll Learn

Demonstrate

Watch an evidence-led example.

Investigate

Work with the supplied synthetic captures and templates.

Verify

Reproduce the observation, then discuss its limitations.

01 Introduction to Network Analysis and Wireshark

A capture is a measurement at one observation point, not a complete network history.

Lab 01 Establish a trace baseline

Use the self-contained lab folder and Learner Guide. Observe → explain → verify → discuss.

Wireshark Structure

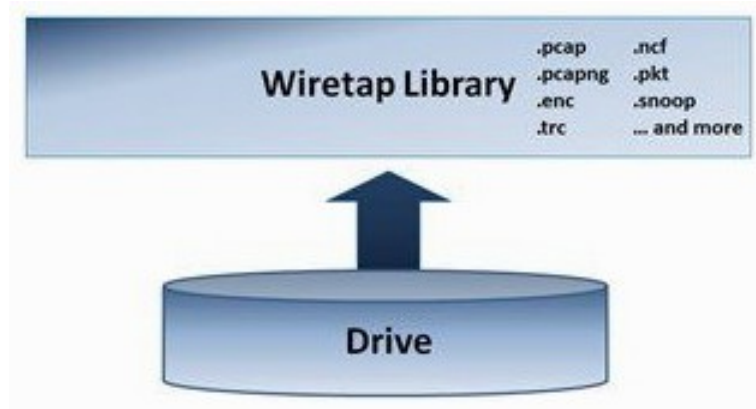
What is Wireshark

- Wireshark is a network packet analyzer. A network packet analyzer presents captured packet data in as much detail as possible.
- You could think of a network packet analyzer as a measuring device for examining what's happening inside a network cable, just like an electrician uses a voltmeter for examining what's happening inside an electric cable (but at a higher level, of course).
- In the past, such tools were either very expensive, proprietary, or both. However, with the advent of Wireshark, that has changed. Wireshark is available for free, is open source, and is one of the best packet analyzers available today.

Why Use Wireshark

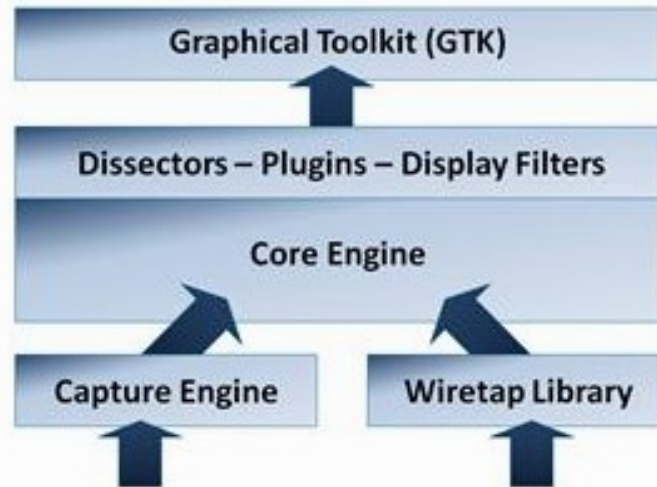
- Here are some reasons people use Wireshark:
 - Network administrators use it to troubleshoot network problems
 - Network security engineers use it to examine security problems
 - QA engineers use it to verify network applications
 - Developers use it to debug protocol implementations
 - People use it to learn network protocol internals

Wiretap Library



- Wiretap Library enables Wireshark to read a variety of trace file formats

Packet Processing Elements in Wireshark



- For example: Dissectors decode packets to display field contents and interpreted values. More on page 41.

Let's Install Wireshark

<https://www.wireshark.org/download.html>

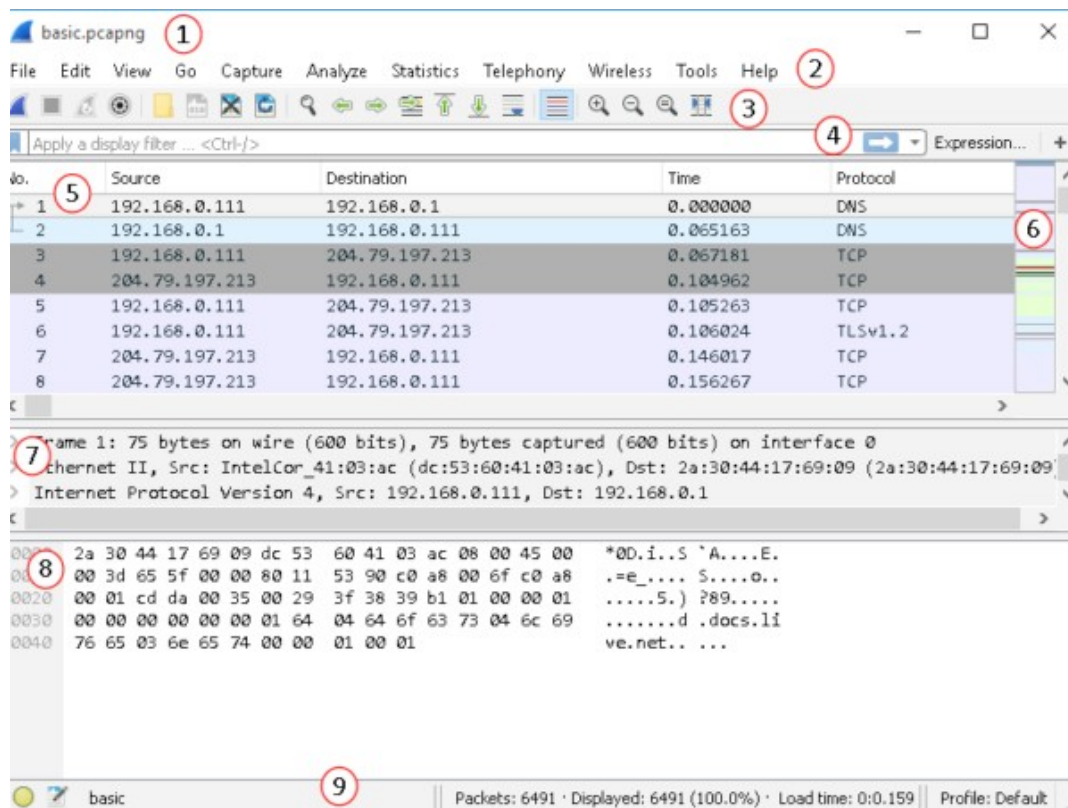
Doc Ref: https://www.wireshark.org/docs/wsug_html/

Wireshark's GUI

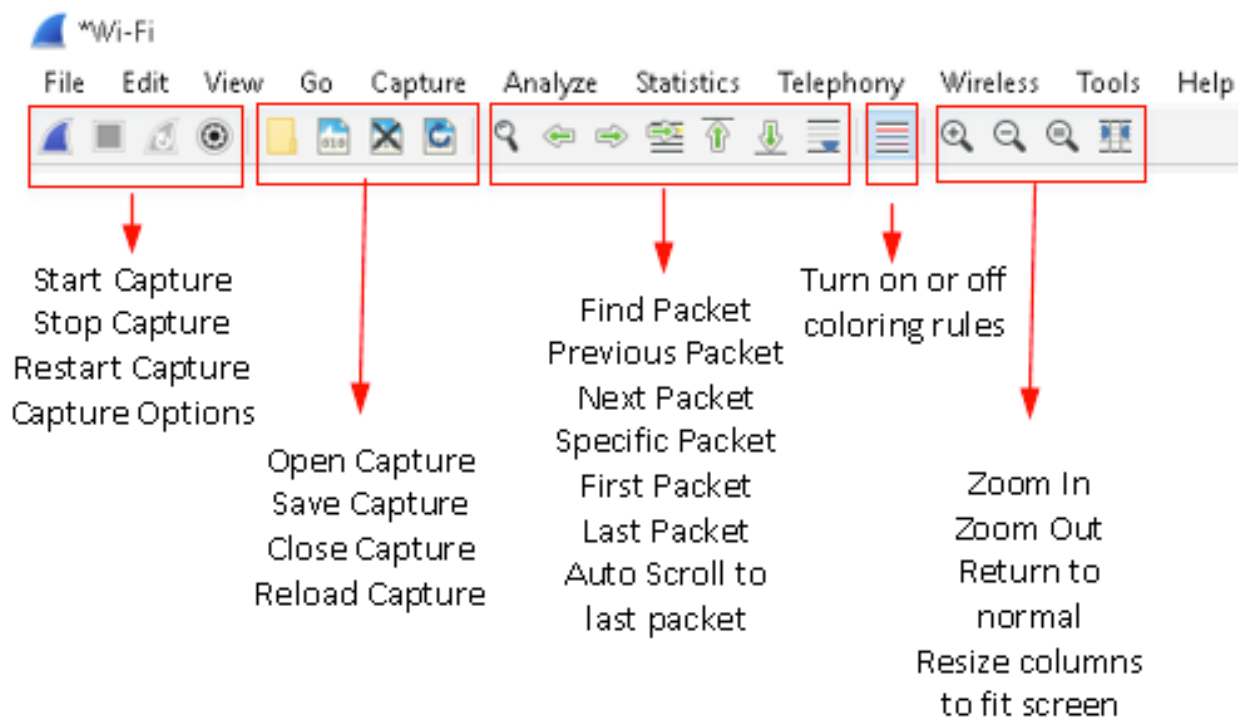
Let's Get Comfortable with

Wireshark Overview

1. Title Bar
2. Main Menu
3. Main Toolbar
4. Filter Toolbar
5. Packet List
6. Intelligent Scrollbar
7. Packet Details
8. Packet Bytes
9. Status Bar



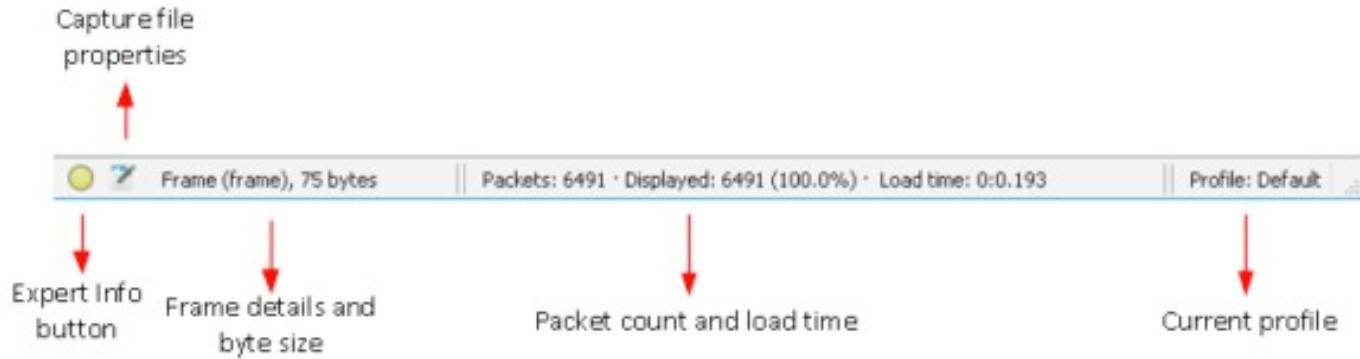
Main Toolbar



Display Filter Toolbar



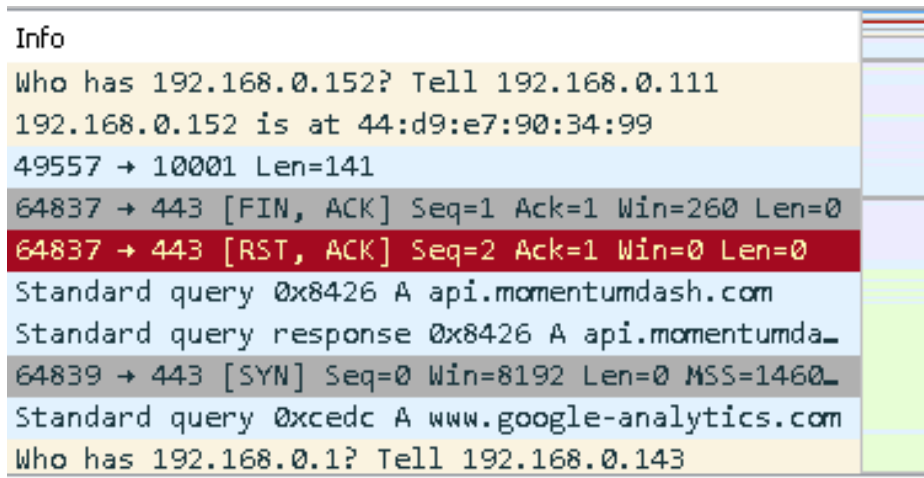
Status Bar



Intelligent Scrollbar

The line will match the color of the packet it corresponds to this will make it easy to spot certain traffic.

For example you could look for red in the scrollbar, this would mean your capture has some tcp resets, aborts or ttl low packets in it.



The image shows a Wireshark packet capture list. On the right side, there is a vertical scrollbar with a color-coded track. The track has segments of yellow, light blue, red, light green, and light blue, corresponding to the packet list items. The packet list items are as follows:

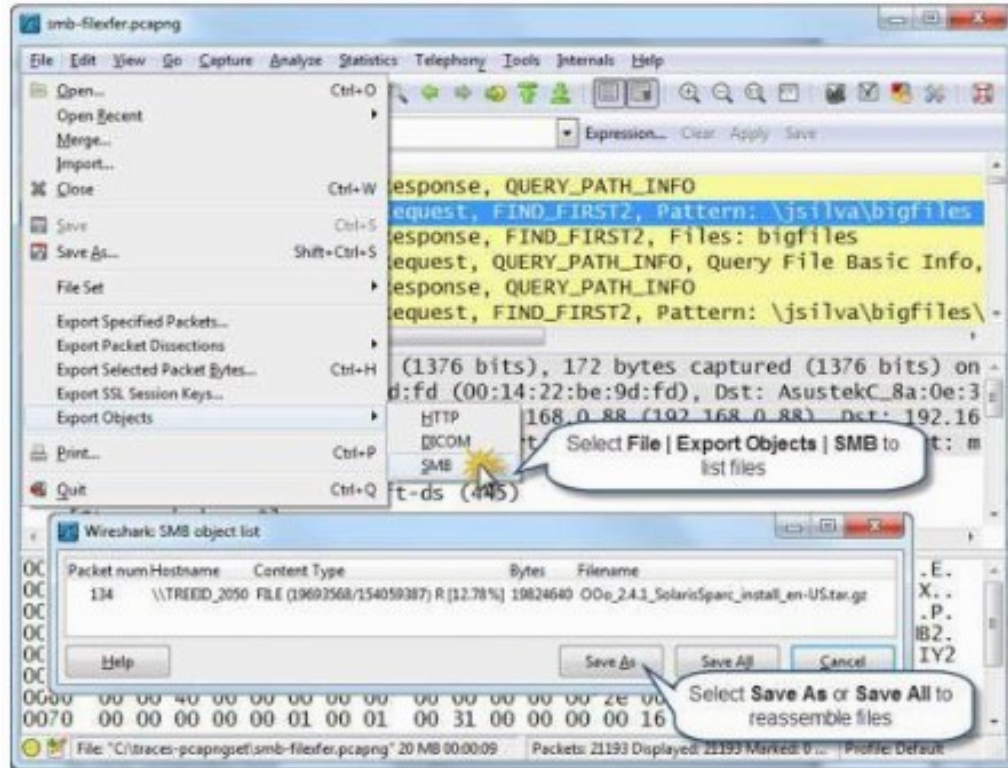
Info
Who has 192.168.0.152? Tell 192.168.0.111
192.168.0.152 is at 44:d9:e7:90:34:99
49557 → 10001 Len=141
64837 → 443 [FIN, ACK] Seq=1 Ack=1 Win=260 Len=0
64837 → 443 [RST, ACK] Seq=2 Ack=1 Win=0 Len=0
Standard query 0x8426 A api.momentumdash.com
Standard query response 0x8426 A api.momentumda_
64839 → 443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460_
Standard query 0xcdec A www.google-analytics.com
Who has 192.168.0.1? Tell 192.168.0.143

Main Menu

The main menu is used to start actions. It consists of the following items:

- **File** – Used to open and merge capture files, save, print or export capture files and close the application.
- **Edit** – Find a packet, mark packets, set time reference, add comment to packets, access configuration profiles and preferences.
- **View** – Controls turning on or off certain windows, change time display format, zoom, access coloring rules, turn on or off colorizing packet list, reset layout and resize columns.
- **Go** – Provides options for moving to a specific packet
- **Capture** – Provides access to start and stop captures and to edit capture files
- **Analyze** – Provides ways to alter displayed packets, turn on or off the dissection of protocols and follow TCP/UDP streams.
- **Statistics** – Provides items to display various statistic windows, such as summary of packets, conversations, I/O graph, capture file properties, and various stats for specific protocols.
- **Telephony** – Provides items to display various details about telephony traffic such as RTP, RCTP, SIP flows, SIP statistics and more.
- **Wireless** – Provides items to show bluetooth and 802.11 wireless statistics.
- **Tools** – Contains a tool to create Firewall ACL rules
- **Help** – Access to manual pages, website, FAQ's, Forum, downloads, wiki and about section.

Exporting SMB



Trace File: **smb-filexfer.pcapng**

Server Message Block
File sharing..

Introduction to Network Analysis and Wireshark

Observe

A capture is a measurement at one observation point, not a complete network history.

Interpret

Packet list, protocol details and bytes connect a summary to the underlying evidence.

Corroborate

Record interface, timestamp precision, dropped packets and capture scope before interpreting a trace.

Lab 01 • Establish a trace baseline

01

Build

A reproducible packet table and written findings.

02

Evidence

2 frame(s) match the core question in the supplied synthetic fixture.

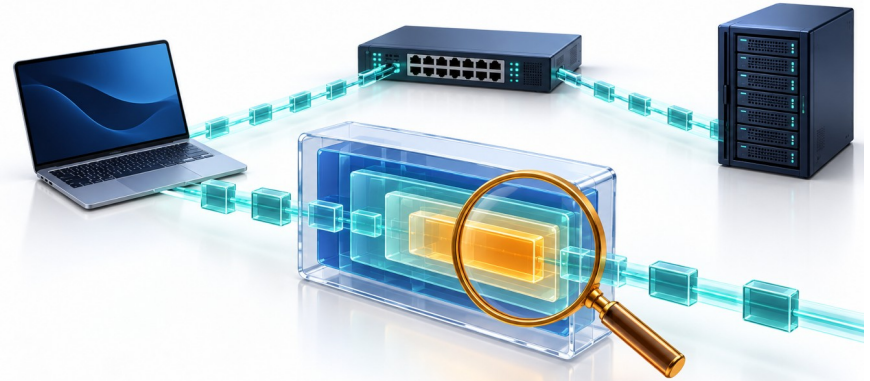
03

Reflect

Which second observation would strengthen your conclusion?

02 Capture Methods and Capture Filters

A switched access port normally observes its own traffic and broadcasts; SPAN or TAP placement changes visibility.



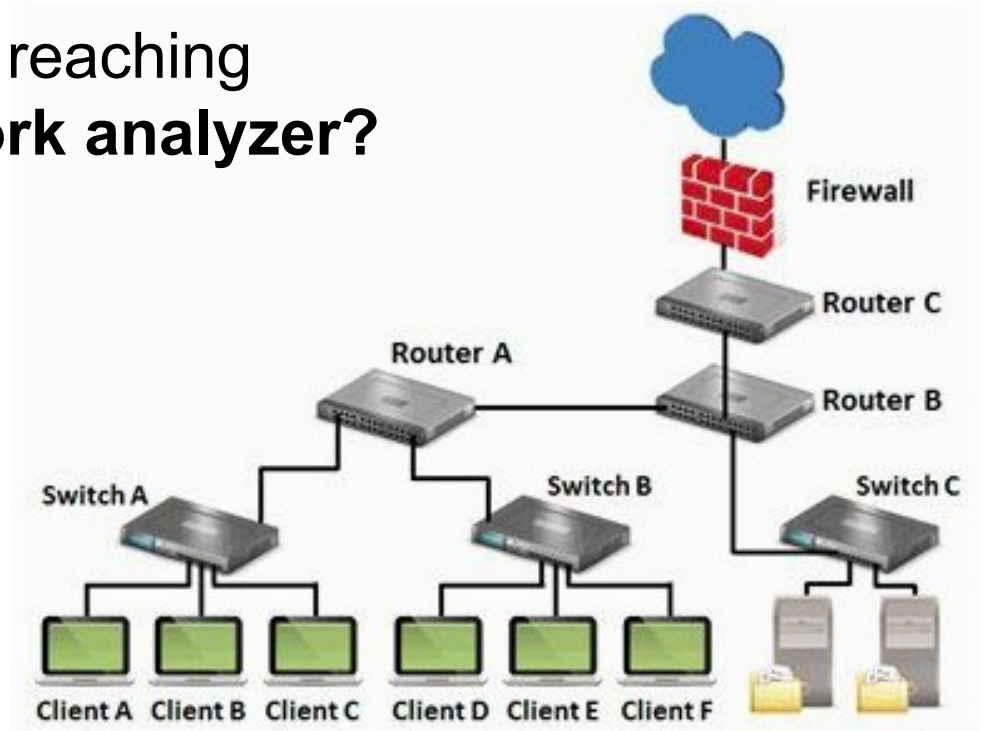
Lab 02 • Plan capture placement

Capture Traffic

Where to Tap Into the Network

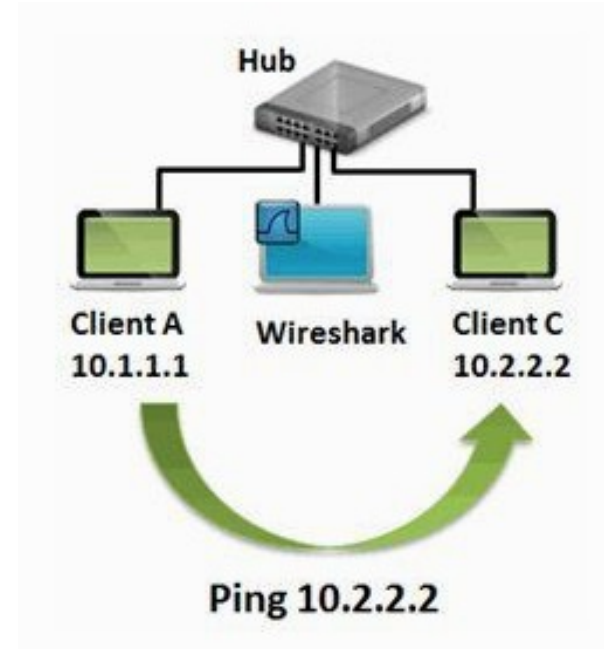
Client A is complaining about reaching Servers, **where to tap network analyzer?**

What if we connect to Switch A?
What type of traffics we can capture?



1. Hub in Half-duplex Network

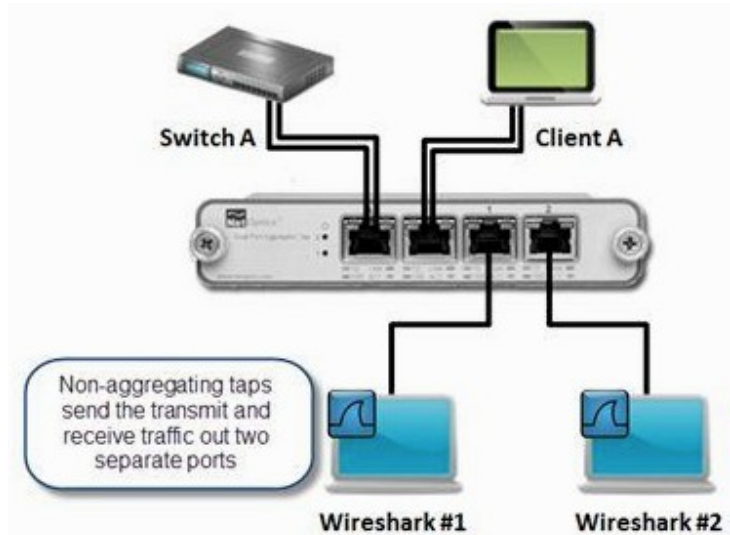
The hub in-line between half-duplex devices.



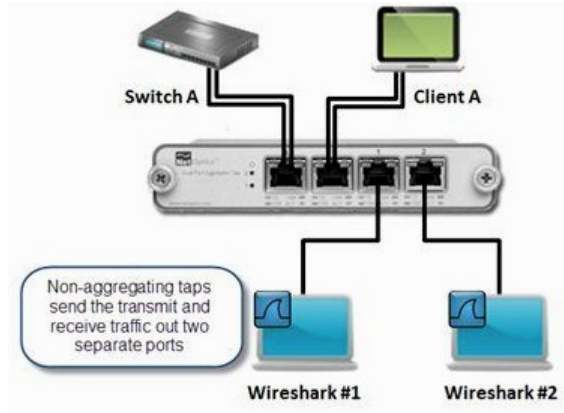
Testing before using

2. Tap into Half/ Full-duplex Traffic

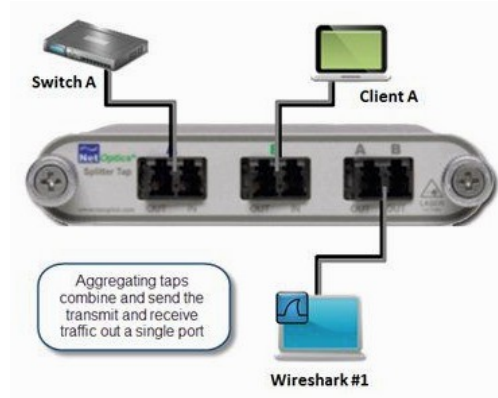
- Use a Test Access Port (TAP) on Half/Full-duplex networks.
- Taps are passive devices that are placed in-line (in the path) between devices.
- Unlike spanned switch ports, taps can forward packets that contain physical layer errors (such as CRC errors) to the monitor port(s).



Aggregating vs. Non-aggregating TAPs

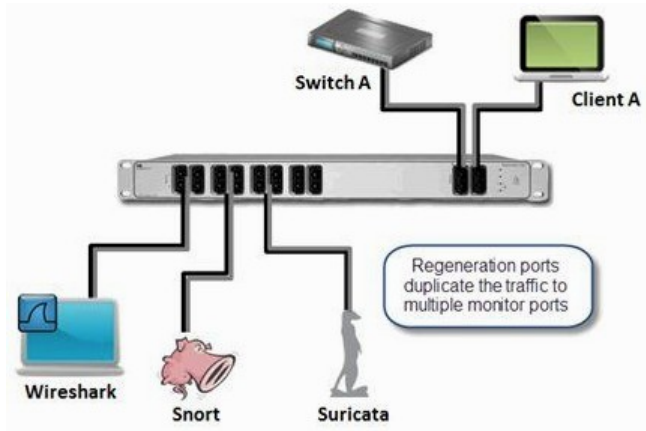


Non-aggregating TAP
Two Ports

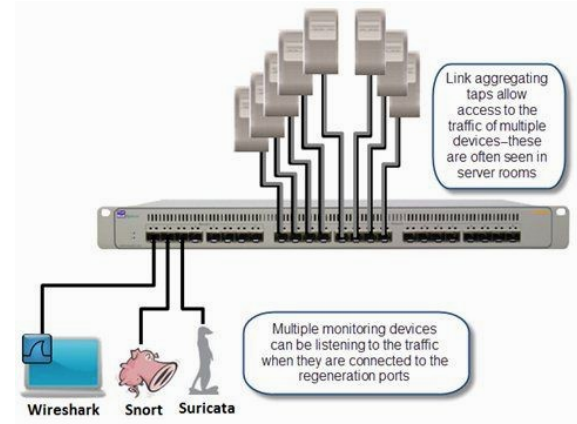


Aggregating TAP
One Ports

Regenerating vs. Link Aggregating TAPs

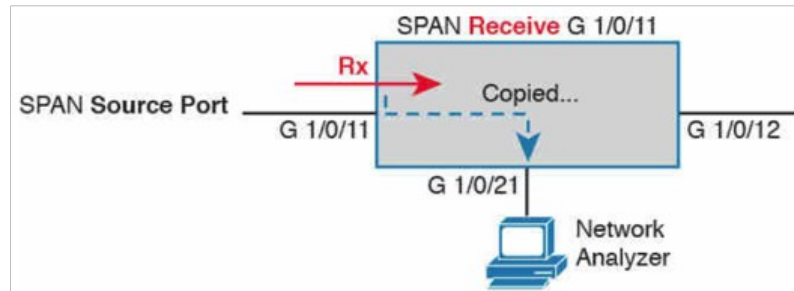
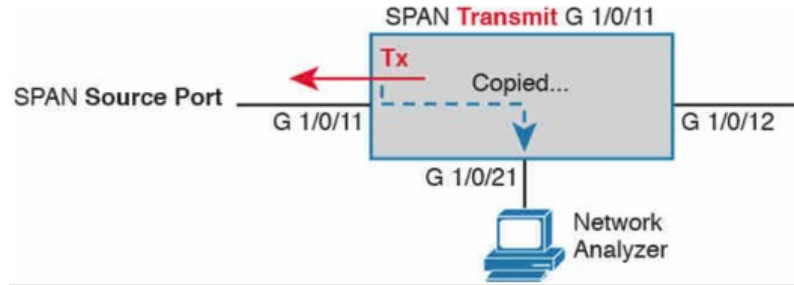


Regenerating TAP
Several Ports



Link Aggregating TAP
Multiple Links to Several Ports

3. Span a Switch Port

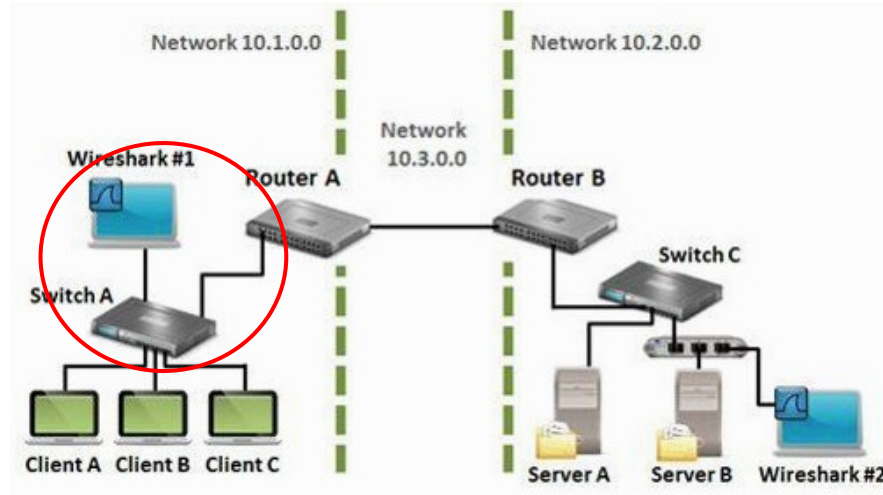


monitor session 1 source interface g1/0/11 {rx | tx | both}

monitor session 1 destination interface g1/0/21

Analyze Routed Networks

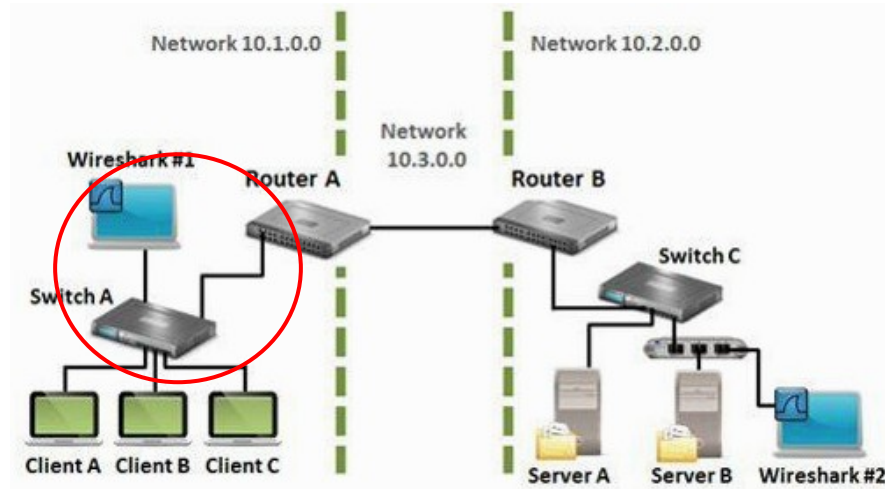
Analyzed Routed Networks



Wireshark #1 is configured through port spanning to listen to the port connecting to Router A.

It means what traffics able to listen?

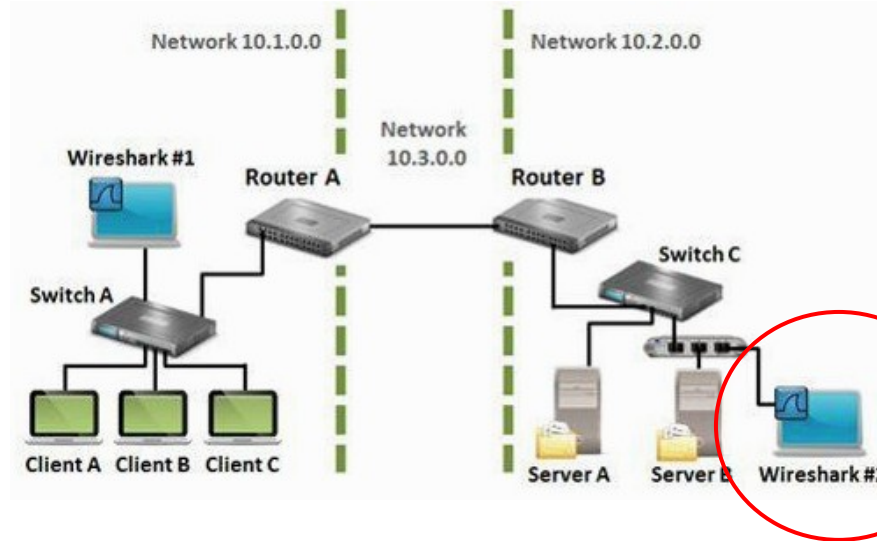
Analyzed Routed Networks



Wireshark #1 is configured through port spanning to listen to the port connecting to Router A.

Enables **Wireshark #1** to listen in on traffic to and from Clients A, B and C and network 10.2.0.0.

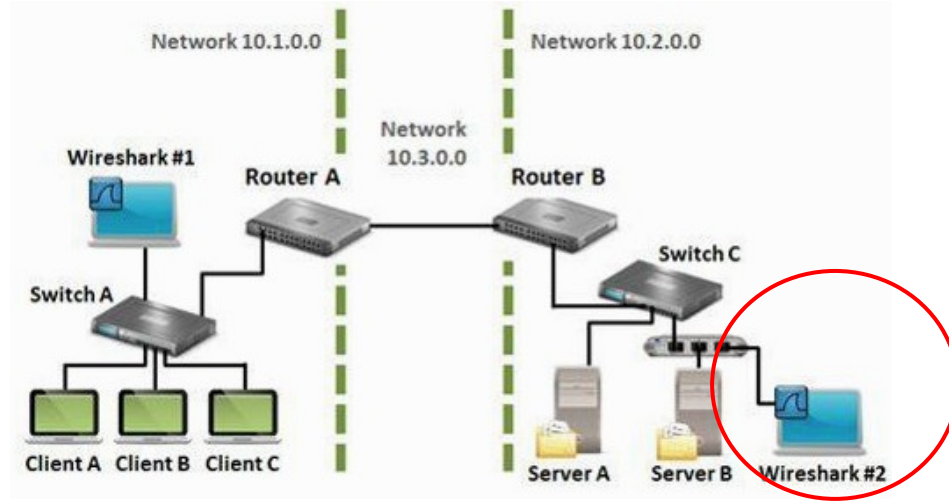
Analyzed Routed Networks



Wireshark #2 is connected to an aggregating tap that connects Server B to Switch C.

It means what traffics able to listen?

Analyze Routed Networks



Wireshark #2 is connected to an aggregating tap that connects Server B to Switch C.

Enables **Wireshark #2** only be able to see traffic to and from Server B and the local and remote networks.

Analyze Wireless Networks

Capture Methods and Capture Filters

01

Observe

A switched access port normally observes its own traffic and broadcasts; SPAN or TAP placement changes visibility.

02

Explain

Capture filters use libpcap syntax before packets are stored; display filters hide or show stored packets.

03

Limit

Bound file size and duration; check capture drops before attributing missing packets to network loss.

Capture Traffic Remotely

Capture Methods and Capture Filters

Observe

A switched access port normally observes its own traffic and broadcasts; SPAN or TAP placement changes visibility.

Interpret

Capture filters use libpcap syntax before packets are stored; display filters hide or show stored packets.

Corroborate

Bound file size and duration; check capture drops before attributing missing packets to network loss.

Lab 02 • Plan capture placement

01

Build

A reproducible packet table and written findings.

02

Evidence

37 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

03 Global Preferences and Troubleshooting Profiles

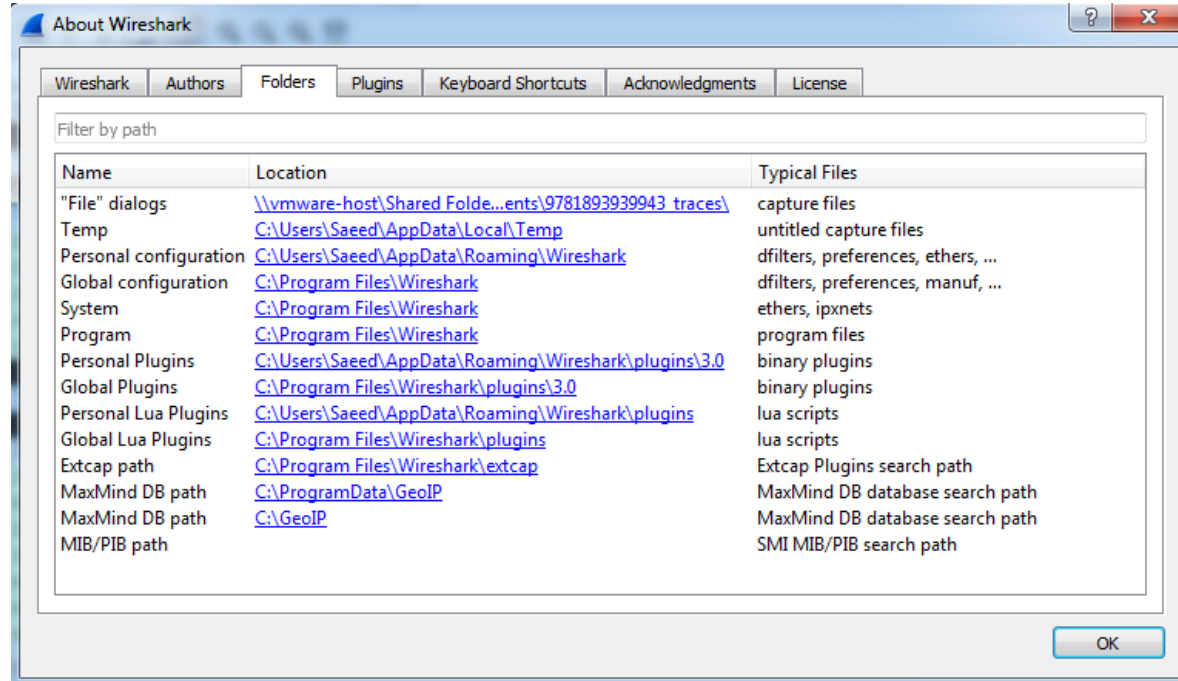
A profile bundles reproducible columns, coloring rules and protocol settings.

Lab 03 Create an analyst profile

Use the self-contained lab folder and Learner Guide. Observe → explain → verify → discuss.

Global and Personal Preferences

Configuration Folders



Help | About Wireshark | Folders

Global Preferences and Troubleshooting Profiles

Observe

A profile bundles reproducible columns, coloring rules and protocol settings.

Interpret

Name resolution can obscure numeric evidence and generate additional traffic; document the setting.

Corroborate

Protocol heuristics and TCP analysis preferences change interpretation, not the stored bytes.

Lab 03 • Create an analyst profile

01

Build

A reproducible packet table and written findings.

02

Evidence

6 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

04 Navigation and Coloring Techniques

Color rules are applied in order; the first matching rule wins.

Lab 04 **Color and annotate evidence**

Use the self-contained lab folder and Learner Guide. Observe → explain → verify → discuss.

Colorize Traffic

Identify Why a Packet is a Certain Color

```
Frame 576: 973 bytes on wire (7784 bits), 973 bytes captured (7784 bits) on int
  Interface id: 0 (unknown)
  Encapsulation type: Ethernet (1)
  Arrival Time: May  4, 2011 22:06:52.219663000 Pacific Daylight Time
  [Time shift for this packet: 0.000000000 seconds]
  Epoch Time: 1304572012.219663000 seconds
  [Time delta from previous captured frame: 0.008882000 seconds]
  [Time delta from previous displayed frame: 0.000000000 seconds]
  [Time since reference or first frame: 0.975825000 seconds]
  Frame Number: 576
  Frame Length: 973 bytes (7784 bits)
  Capture Length: 973 bytes (7784 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ip:tcp:http:png]
  [Coloring Rule Name: New coloring rule]
  [Coloring Rule String: tcp.analysis.flags && !tcp.analysis.window_update]
  Ethernet II, Src: Cadent 31:kh:c1:00:01:5c:31:kh:c1, Dst: Hewlett-P 07:bf:a3: /
```

Tip/Key Point

What is the difference between marking packets and applying a coloring rule?

Tip/Key Point

What is the difference between marking packets and applying a coloring rule?

Packet marking is a temporary designation that is cleared when you reload the trace file, open the trace file again or toggle the packet marking off.

Coloring rules are automatically applied to the traffic each time you open the trace file (if coloring is enabled).

Tip/Key Point

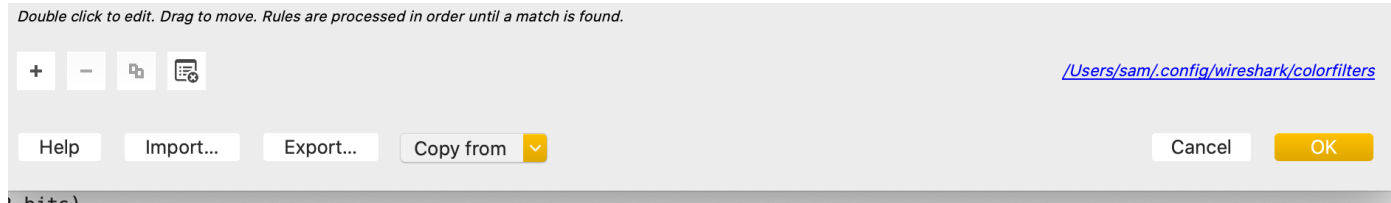
How do you share coloring rules with other Wireshark users?

Tip/Key Point

How do you share coloring rules with other Wireshark users?

By default, coloring rules are contained in the colorfilters file. This file can be copied to another Wireshark system.

In addition, you can use the export and import feature in the coloring rules window to save the coloring rules file by another name and load it on another Wireshark system.



Navigation and Coloring Techniques

Observe

Color rules are applied in order; the first matching rule wins.

Interpret

Temporary coloring is useful for a conversation; permanent rules support repeated triage.

Corroborate

Bookmarks and comments preserve the analyst path without changing the captured payload.

Lab 04 • Color and annotate evidence

01

Build

A reproducible packet table and written findings.

02

Evidence

2 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

05 Time Values and Delay Types

Displayed delta depends on the current filter; capture delta does not.

Lab 05 **Separate path and server delay**

Use the self-contained lab folder and Learner Guide. Observe → explain → verify → discuss.

What Triggers **Window Update**?

These are actually good packets. A client just advertised a larger receive buffer space indicating an application just picked up some data from the receive buffer.

TCP Previous segment not captured

- If packet N is marked with previous segment not captured, it means that in the capture there is no packet from the same TCP session whose seq + length would match the seq of packet N.
- The most typical reason is packet loss and/or late start of capture..
- But there can eventually be other reasons (buggy TCP stack of the sender, multipath network structure allowing packets belonging to the same TCP session to pass through different network interfaces so the packets do reach their destination but Wireshark cannot see them, ...), so it is up to you to check out the real reason why this has happened in your particular case.

Define Time Values and Interpret Summaries

Use Time to Identify Network Problems

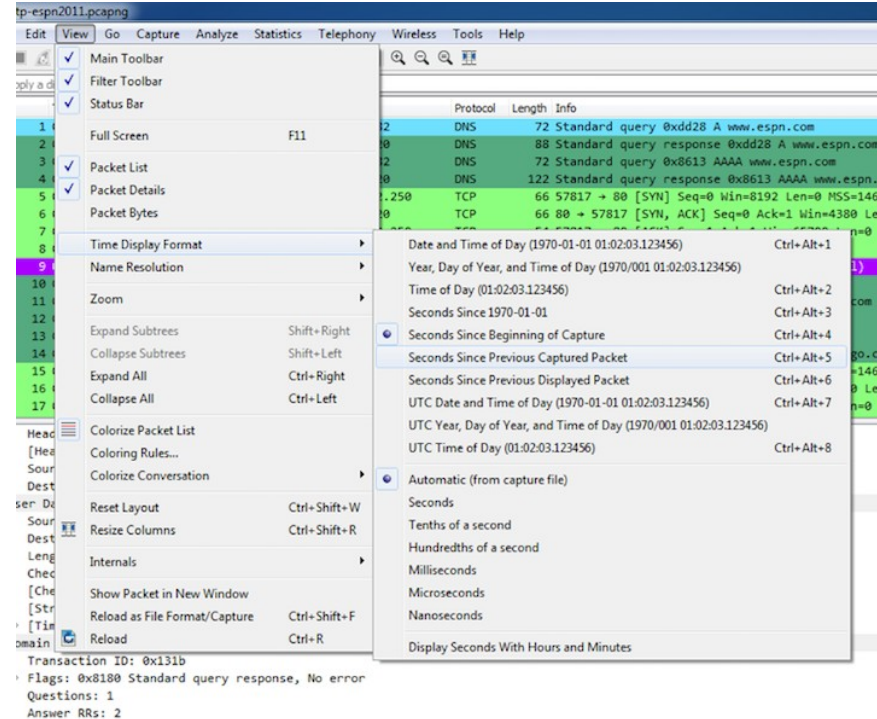
- When troubleshooting **slow network** communications, it is important to focus on the Time column.
- Slow network performance can be due to
 - high latency,
 - access errors,
 - excessive number of packets required to obtain data or a number of other causes.
- When poor performance is due to delays in the communications, look for
 - large gaps in time between a request and acknowledgement, an acknowledgement and a response, etc.

Choose the Ideal Time Display Format

seconds microseconds

32.130943090

milliseconds nanoseconds



Seconds since [Previous Captured Packet] vs. [Previous Displayed Packet]

- **Seconds since Previous Captured Packet**
 - often **called** the delta time setting
 - measures the time from the end of one packet to the end of the next packet.
 - If a display filter is set and a packet is not displayed, its **timestamp is still calculated** and shown on packets that are displayed.
- **Seconds since Previous Displayed Packet**
 - only counts the delta time value from the end of one displayed packet to the next displayed packet.

	Captured Packet	Displayed Packet
1	0.000000	0.000000
2	0.001000	0.001000
3	0.013000	-----
4	0.002000	0.015000
5	0.034000	-----
6	0.006000	-----
7	0.018000	0.058000
8	0.005000	0.005000

How can the time setting be used to identify the cause of network performance problems?

How can the time setting be used to identify the cause of network performance problems?

One way to identify network problems is to set the Time column to Seconds since Previously Displayed Packet and look for large gaps in time in a conversation during what should be an automated streaming process.

For example, during a file transfer process the file should be transferred without large gaps in time.

Time Values and Delay Types

Observe

Displayed delta depends on the current filter; capture delta does not.

Interpret

SYN to SYN/ACK gives an initial path-related sample; request to response includes application processing.

Corroborate

One-sided timestamps cannot identify exactly which intermediate device delayed or dropped a packet.

Lab 05 • Separate path and server delay

01

Build

A reproducible packet table and written findings.

02

Evidence

3 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

06 Trace Statistics and VoIP Overview

Protocol Hierarchy shows captured composition; byte share differs from packet share.

Lab 06

Summarize traffic and a voice stream

Use the self-contained lab folder and Learner Guide. Observe → explain → verify → discuss.

You have opened a trace file sent to you from another company. The timestamp only shows millisecond resolution. Why? Can you improve the timestamp resolution of the trace file?

You have opened a trace file sent to you from another company. The timestamp only shows millisecond resolution. Why? Can you improve the timestamp resolution of the trace file?

Most likely the analyzer used to capture the trace file could not provide more precise timestamps. You cannot alter the timestamp resolution of captured trace files.

You have opened a trace file that contains 5 separate conversations. How can Time Reference be used to measure the time elapsed in one of the conversations?

Tip/Key point

You have opened a trace file that contains 5 separate conversations. How can Time Reference be used to measure the time elapsed in one of the conversations?

You could set a Time Reference on the first packet of the conversation you are interested in and scroll to the end of the conversation.

The Time column will indicate the time elapsed from the Time Reference packet and the last packet of the conversation.

Alternately you could filter on the conversation of interest and then set the Time Reference on the first packet. The last packet displayed indicates the time elapsed for the conversation.

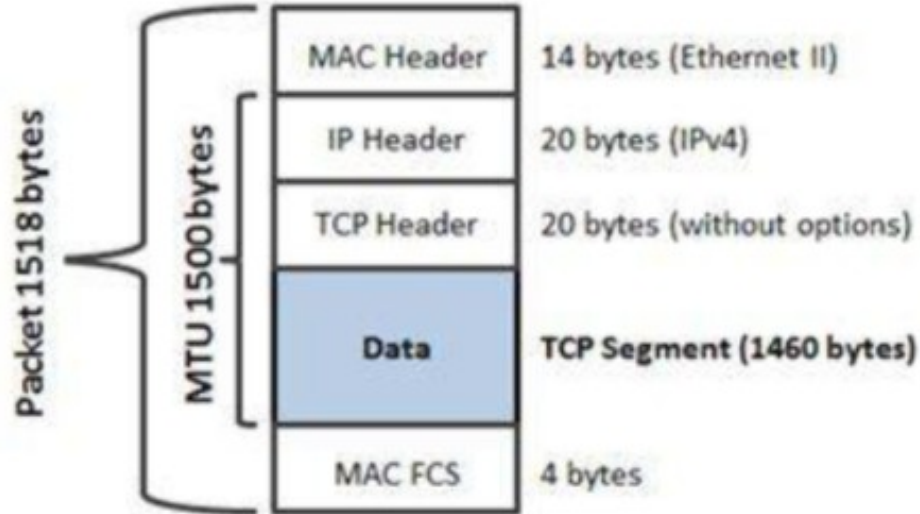
Interpret Basic Trace File Statistics

Evaluate Packet Lengths

Maximum Transmission Unit (MTU), etc.

Evaluate Packet Lengths

- The most common network file transfer methods use TCP/IP communications over Ethernet.
- Standard TCP packet structure as follow:



What is the problem of using small packet size to transfer a large amount of data?

Use Flow Graphs to Spot Web Browsing Issues

- Creating Flow Graphs of web browsing sessions displays the number of sites you are redirected to for content when accessing a site.
- For example, to load www.espn.com's main web page, the Flow Graph indicates that a client must connect to numerous web servers.
- If one of these other web servers responds slowly or with 404 errors, the user may complain about slow web browsing or state that part of the site does not load.

How can you use the Protocol Hierarchy window to identify a breached host?

How can you use the Protocol Hierarchy window to identify a breached host?

After capturing traffic to and from the host, open the Protocol Hierarchy window to look for unusual applications such as TFTP, IRC, etc.

You can apply a display filter for the conversation from inside the Protocol Hierarchy window and then follow the TCP or UDP stream to reassemble the communications and identify commands or information exchanged.

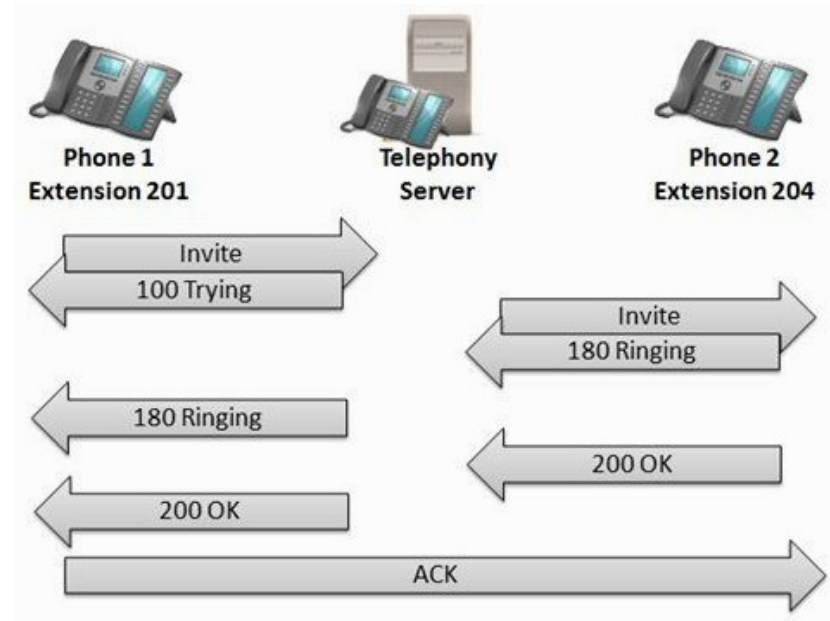
Your trace file contains over 100 TCP connections. How can you identify the most active TCP connection based on bytes per second?

Voice over IP (VoIP) Analysis

Voice over IP (VoIP)

VoIP communications consist of two primary parts

1. The signaling protocol for call setup and teardown and
 - a. Session Initiation Protocol (SIP)
 - b. SIP runs over UDP or TCP port 5060 (UDP common)
2. the transport protocol for the voice communications.
 - a. Real-time Transport Protocol (RTP)
 - b. RTP runs over any even port number



Trace Statistics and VoIP Overview

Observe

Protocol Hierarchy shows captured composition; byte share differs from packet share.

Interpret

Conversations and Endpoints identify concentration; they do not alone prove malicious activity.

Corroborate

SIP signals a call while RTP carries media; jitter, loss and codec interpretation need stream context.

Lab 06 • Summarize traffic and a voice stream

01

Build

A reproducible packet table and written findings.

02

Evidence

2 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

07 Display Filters

Parentheses make mixed and/or expressions explicit.

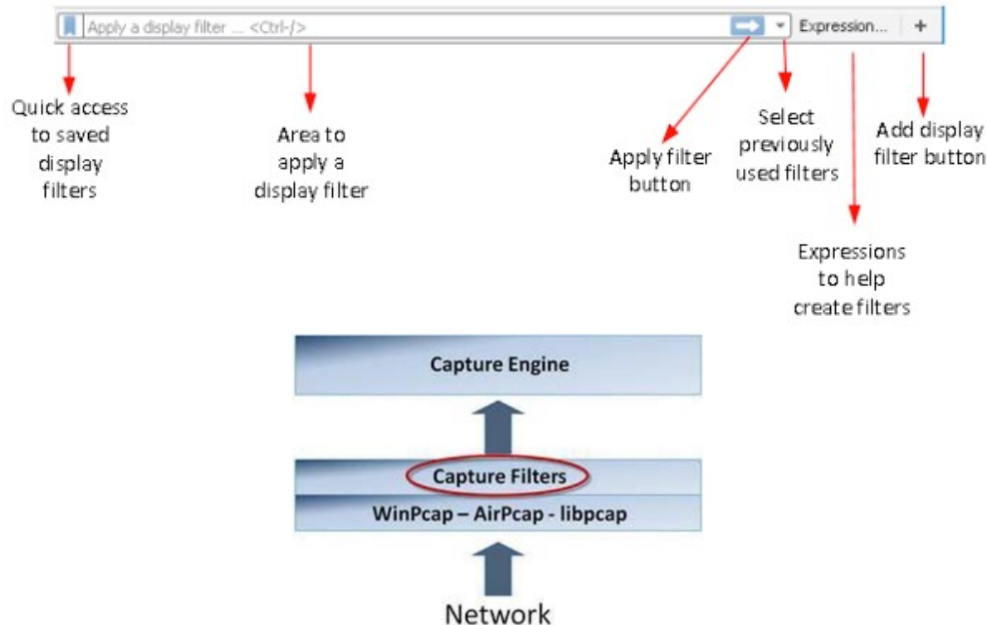
Lab 07 **Build a filter evidence matrix**

Use the self-contained lab folder and Learner Guide. Observe → explain → verify → discuss.

Create and Apply Capture Filters

Capture Filter

- Capture filters use the Berkeley Packet Filtering (BPF) syntax, the same filter syntax used by tcpdump.



BPF Syntax

Identifiers

- In a capture filter for traffic to or from port 53, "53" is the identifier.
- The identifier can be a decimal or hexadecimal number or an ASCII string.

Qualifiers

1. Type
 - a. The type of name or number to which the identifier refers
 - b. Port, host, net, ... e.g. port 53
2. Dir
 - a. The direction such as src, dst.
3. Proto
 - a. The protocol qualifier such as TCP, UDP.

`ip.src == 199.181.132.250`

Visit: <http://biot.com/capstats/bpf.html>

Filter Comparison Operators

Shortcut	C-Like	Description	Example
eq	==	Equal	<code>Ip.src == 10.1.1.5</code>
ne	!=	Not equal	<code>Ip.src != 10.1.1.5</code>
gt	>	Greater than	<code>Frame.len > 64</code>
lt	<	Less Than	<code>Frame.len < 1518</code>
ge	>=	Greater then or equal to	<code>Frame.len ge 0x100</code>
le	<=	Less then or equal to	<code>Frame.len <= 0x20</code>

Display Filters

Observe

Parentheses make mixed and/or expressions explicit.

Interpret

Field existence and Boolean equality differ: `tcp.flags.syn == 1` tests the bit.

Corroborate

Since Wireshark 3.6, `!=` uses all-not-equal semantics; historical slides describing any-not-equal are obsolete.

Lab 07 • Build a filter evidence matrix

01

Build

A reproducible packet table and written findings.

02

Evidence

1 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

08 TCP/IP Communications and Resolution

ARP resolves a local next-hop MAC; DNS resolves a name to an address.



Lab 08 • Reconstruct an application dependency chain

Recap the Basics

Understand Network Traffic Flows

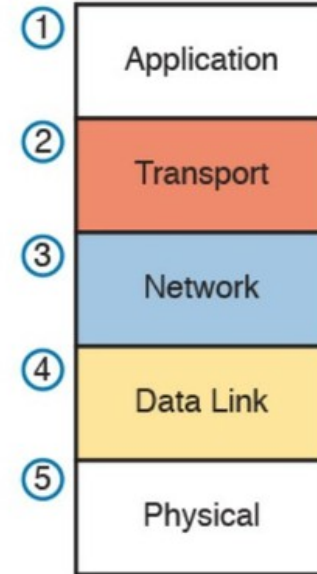
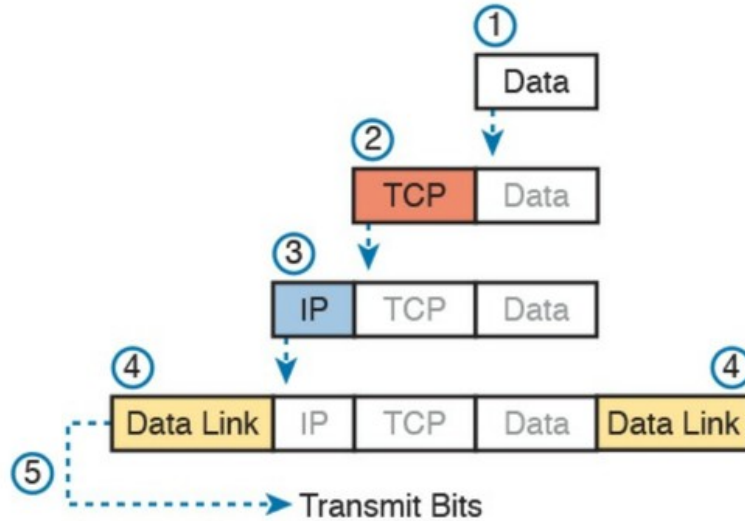
TCP/IP Protocol Suite

TCP/IP v.s OSI Model

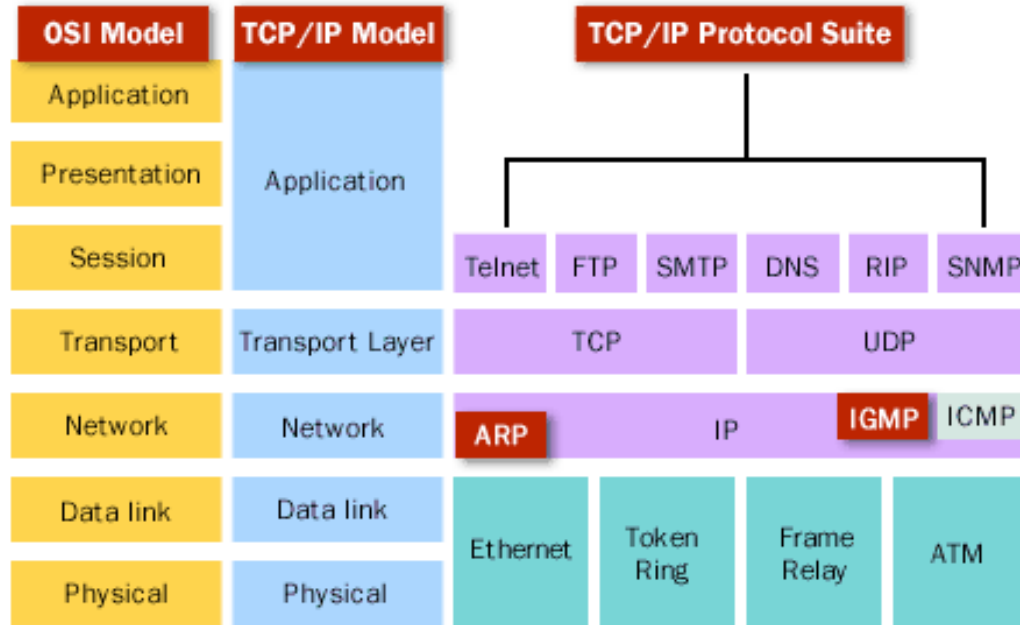
OSI Model	TCP/IP Model
Application	Application
Presentation	
Session	
Transport	Transport Layer
Network	Network
Data link	Data link
Physical	Physical

PDU

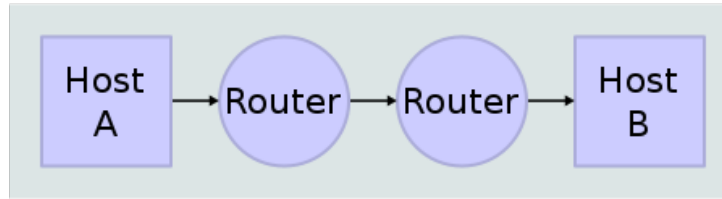
* PDU: Protocol Data Unit



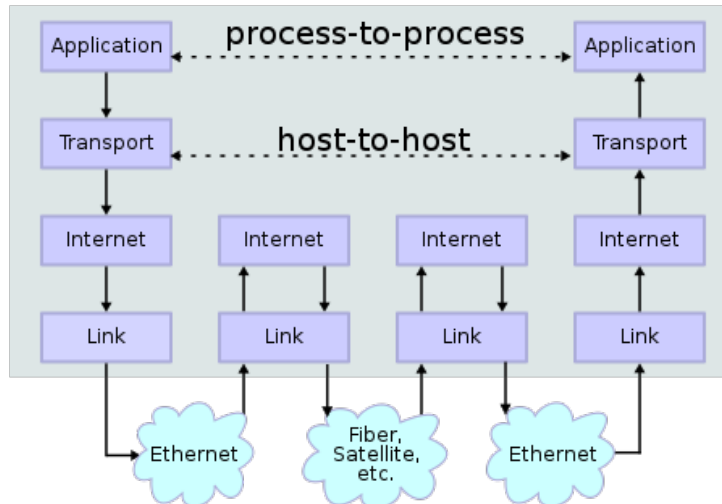
TCP/IP Protocol Suite



Network Topology

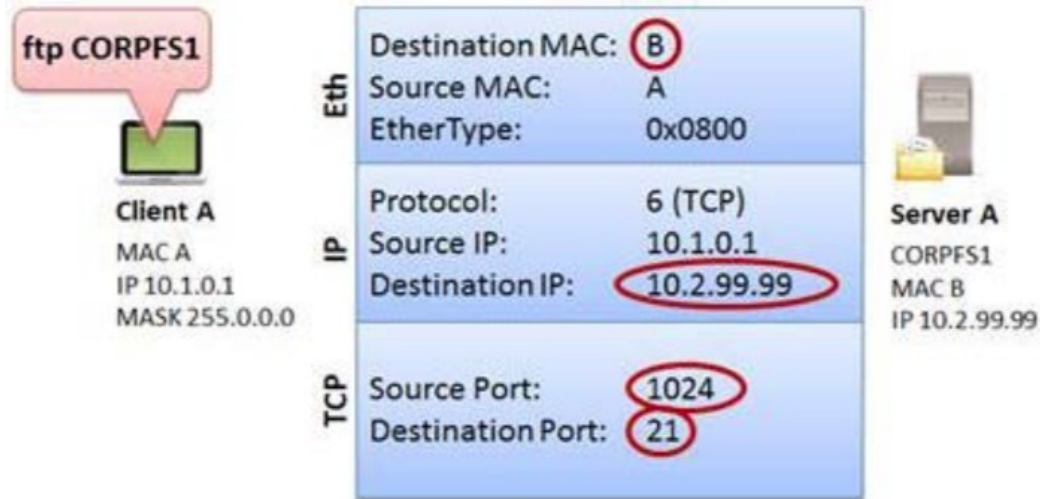


Data Flow



Building a Packet

Assume we are sending packet to a FTP server.



Capture Methods and Capture Filters

01

Observe

A switched access port normally observes its own traffic and broadcasts; SPAN or TAP placement changes visibility.

02

Explain

Capture filters use libpcap syntax before packets are stored; display filters hide or show stored packets.

03

Limit

Bound file size and duration; check capture drops before attributing missing packets to network loss.

ICMP and ARP

ICMP/ICMPv6

- Internet Control Message Protocol (ICMP/ICMPv6) is used to provide network information and is typically recognized as the protocol used for ping.
- **ICMPv6 is used to check and see if an IPv6 address is already in use (duplicate address detection).**

ARP

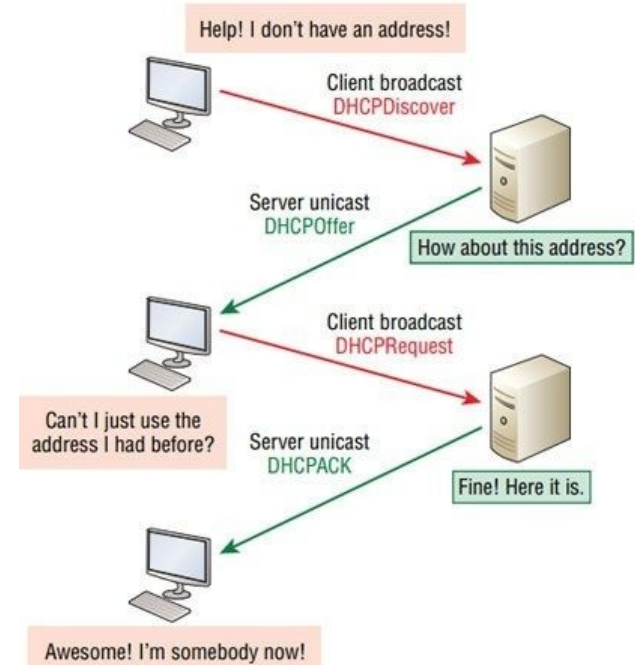
- Address Resolution Protocol (ARP) provides hardware address lookup services for a local device.
- **ARP also enables us to check and see if an IPv4 address is already in use (duplicate address test).**

DHCP

- Dynamic Host Configuration Protocol (DHCP) provides dynamic client configuration and service discovery services.
- DHCP can also provide default gateway settings, DNS server settings and more.

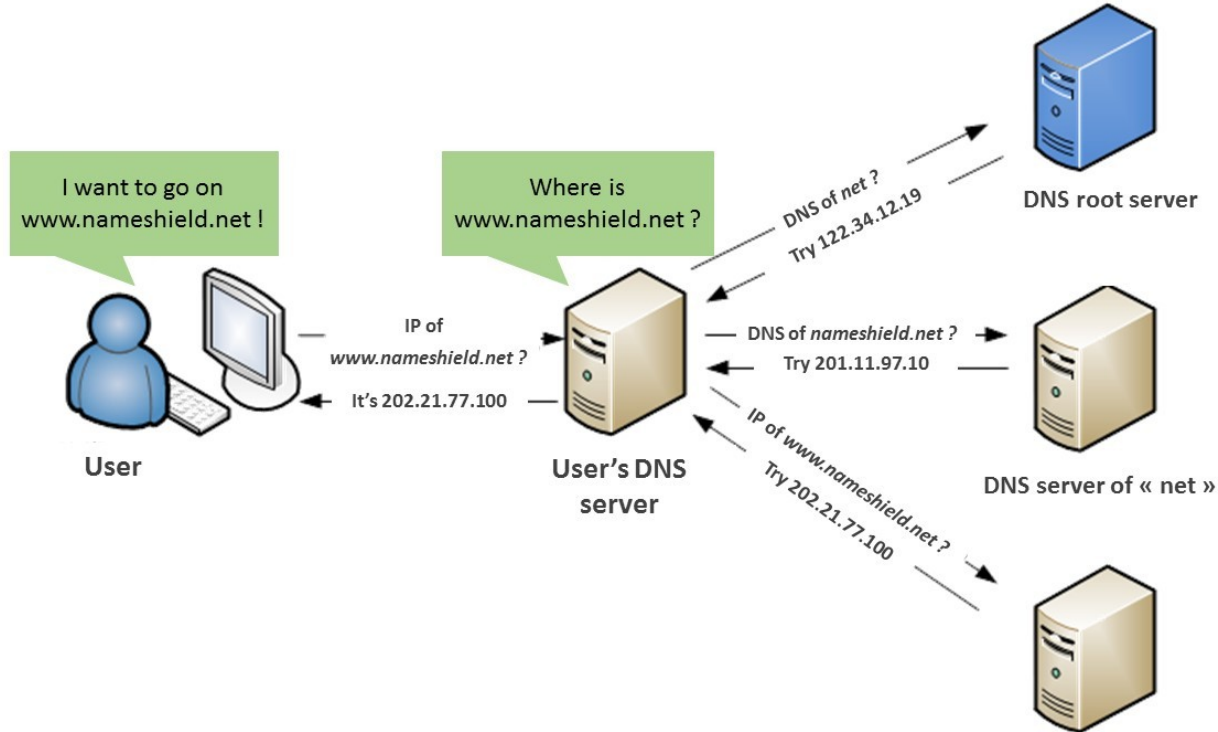
No.	Time	Source	Destination	Protocol	Length	Info
235	22.191805	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x3ce21374
237	22.212148	10.114.30.5	10.114.30.180	DHCP	360	DHCP Offer - Transaction ID 0x3ce21374
238	22.212400	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x3ce21374
239	22.217302	10.114.30.5	10.114.30.180	DHCP	360	DHCP ACK - Transaction ID 0x3ce21374

Frame 235: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface 0
Ethernet II, Src: 10.114.30.180 (00:09:6b:42:ff:ff), Dst: 255.255.255.255
Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
User Datagram Protocol, Src Port: bootpc (68), Dst Port: bootpd (67)
Bootstrap Protocol

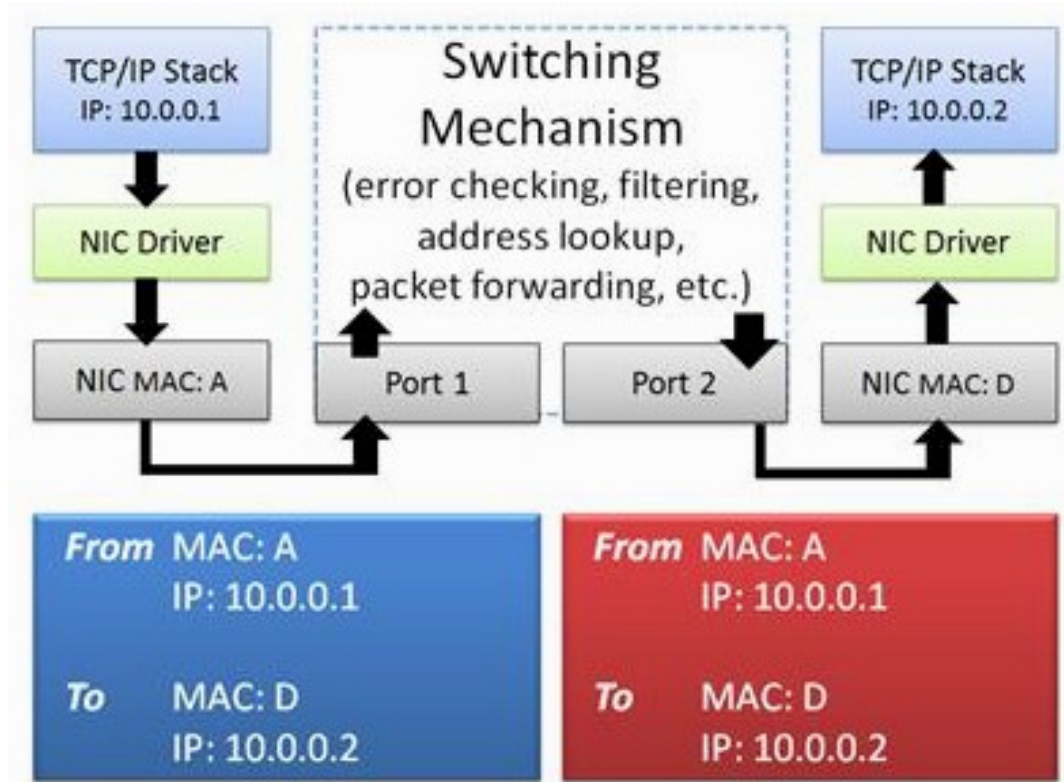


DNS

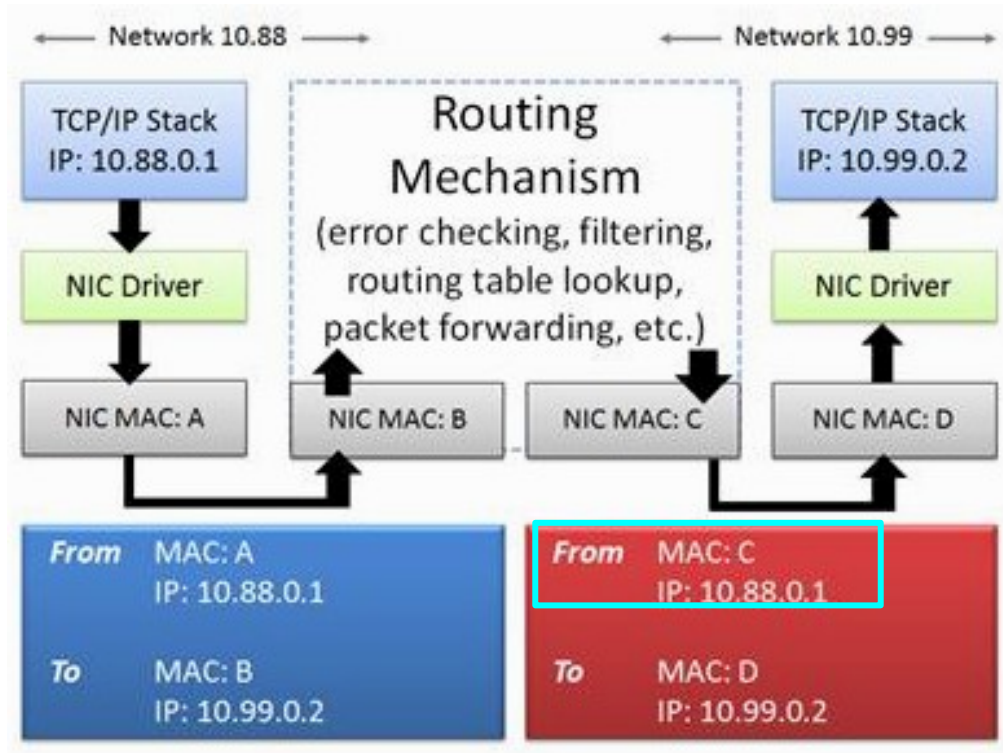
Domain Name System (DNS) provides host name-to-IP address resolution services.



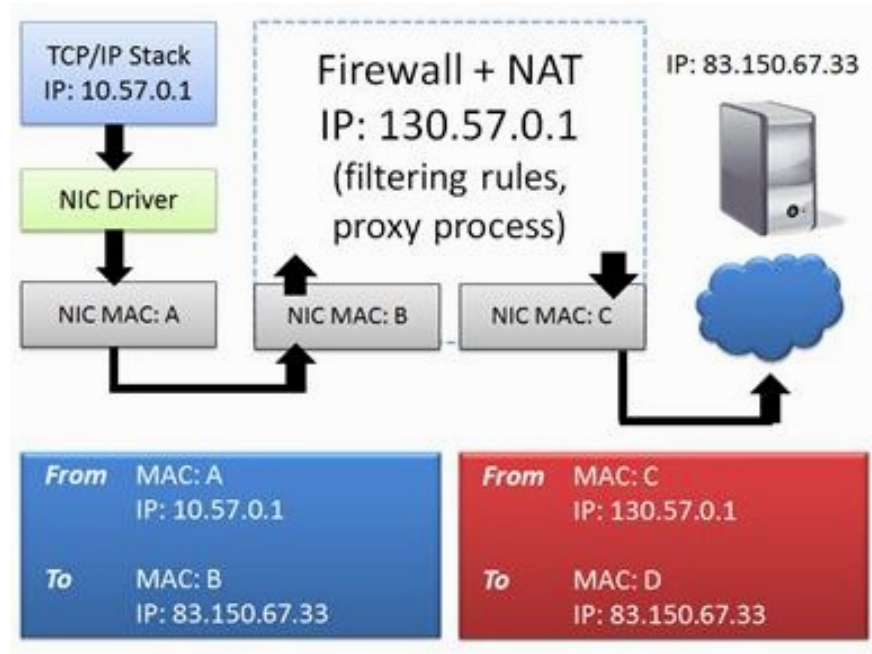
Switching Overview



Routing Overview



Proxy, Firewall and NAT/PAT Overview



Other Technologies that Affect Packets

VLAN Tagging

- Virtual LAN (VLAN) tagging (defined as 802.1Q) adds an identification (tag) to the packets.

MPLS Labeling

- Multiprotocol Label Switching (MPLS) is a method of creating virtual links between remote hosts.

Smart Hardware

- ASA strips off some TCP functionality during the connection process.
- WAN optimization techniques can also alter the packet and data stream process by compressing traffic, offering locally-cached versions of data, optimizing TCP or prioritizing traffic based on defined characteristics (traffic "shaping").

Best Practice: to capture the packets before and after they pass through a traffic-altering device.

Summary

L#	Device Type	OSI Layer	TCP/IP Org.	TCP/IP New	Protocols	PDU
7		Application	Application	Application	HTTP, FTP, POP,SMTP, DNS, RIP	Data
6		Presentation				Data
5		Session				Data
4		Transport	Transport	Transport	TCP/UDP	Segments
3	Router	Network	Internet	Network	IP, ARP, ICMP, IGMP	Packets
2	Switch/Bridge	Data Link	Link	Data Link	Ethernet, Token Ring	Frames
1	Hubs/Repeater	Physical		Physical		Bits

TCP/IP Communications and Resolution

Observe

ARP resolves a local next-hop MAC; DNS resolves a name to an address.

Interpret

The destination IP stays end-to-end across routing while link-layer addresses change per hop.

Corroborate

Resolution, connection establishment and application exchange form a dependency chain.

Lab 08 • Reconstruct an application dependency chain

01

Build

A reproducible packet table and written findings.

02

Evidence

2 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

09 DNS Traffic Analysis

Transaction ID plus addresses and ports pair a query with its response.

Lab 09 Diagnose DNS failure and delay

Use the self-contained lab folder and Learner Guide. Observe → explain → verify → discuss.

DNS Traffic Analysis

Observe

Transaction ID plus addresses and ports pair a query with its response.

Interpret

NXDOMAIN reports that a name does not exist; it differs from silence or timeout.

Corroborate

DNS response time is a measured exchange, influenced by resolver and path behaviour.

Lab 09 • Diagnose DNS failure and delay

01

Build

A reproducible packet table and written findings.

02

Evidence

1 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

10 ARP Traffic Analysis

ARP requests are link-local broadcasts; replies usually return to the requester.

Lab 10 Verify link-local resolution

Use the self-contained lab folder and Learner Guide. Observe → explain → verify → discuss.

ARP Traffic Analysis

Observe

ARP requests are link-local broadcasts; replies usually return to the requester.

Interpret

A repeated unresolved request suggests a local resolution problem, but a single trace may miss the reply.

Corroborate

MAC changes require corroboration before claiming duplicate IP or spoofing.

Lab 10 • Verify link-local resolution

01

Build

A reproducible packet table and written findings.

02

Evidence

1 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

11 IPv4 Traffic Analysis

TTL limits forwarding hops; it is not a latency measurement.

Lab 11 **Classify IPv4 scope and headers**

Use the self-contained lab folder and Learner Guide. Observe → explain → verify → discuss.

IPv4 Traffic Analysis

Observe

TTL limits forwarding hops; it is not a latency measurement.

Interpret

Broadcast and multicast use different addressing scopes and delivery rules.

Corroborate

Fragmentation fields describe packet handling; missing fragments can reflect capture limitations.

Lab 11 • Classify IPv4 scope and headers

01

Build

A reproducible packet table and written findings.

02

Evidence

1 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

12 ICMP Traffic Analysis

Echo requests/replies show an IP exchange; they do not guarantee an application works.

Lab 12 Interpret echo and unreachable messages

Use the self-contained lab folder and Learner Guide. Observe → explain → verify → discuss.

ICMP Traffic Analysis

Observe

Echo requests/replies show an IP exchange; they do not guarantee an application works.

Interpret

Destination unreachable messages quote the offending datagram.

Corroborate

ICMP may be filtered or rate-limited, so absence of a reply needs cautious interpretation.

Lab 12 • Interpret echo and unreachable messages

01

Build

A reproducible packet table and written findings.

02

Evidence

1 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

13 UDP Traffic Analysis

UDP has no transport handshake or retransmission; the application may provide reliability.

Lab 13 **Follow datagrams and service refusal**

Use the self-contained lab folder and Learner Guide. Observe → explain → verify → discuss.

UDP Traffic Analysis

Observe

UDP has no transport handshake or retransmission; the application may provide reliability.

Interpret

A port number is a decoding hint; payload and context identify the application.

Corroborate

Following a UDP stream collects datagrams and does not invent missing data.

Lab 13 • Follow datagrams and service refusal

01

Build

A reproducible packet table and written findings.

02

Evidence

2 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

14 TCP Protocol Analysis

Sequence numbers count bytes; ACK numbers indicate the next expected byte.



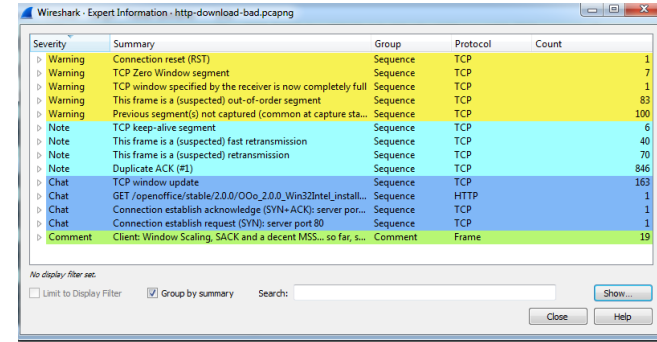
Lab 14 • Investigate retransmission and zero window

Expert System

Expert Information

Expert Information is classified into one of four categories:

- **Errors:** Packet or dissector errors
- **Warnings:** Unusual responses from the application/transport
- **Notes:** Unusual responses from the application/transport, but may not be serious
- **Chats:** Information about the workflow
- **Comment**



Wireshark - Expert Information - http-download-bad.pcapng

Severity	Summary	Group	Protocol	Count
Warning	Connection reset (RST)	Sequence	TCP	1
Warning	TCP Zero Window segment	Sequence	TCP	7
Warning	TCP window specified by the receiver is now completely full	Sequence	TCP	1
Warning	This frame is a (suspected) out-of-order segment	Sequence	TCP	83
Warning	Previous segment(s) not captured (common at capture sta...	Sequence	TCP	100
Note	TCP keep-alive segment	Sequence	TCP	6
Note	This frame is a (suspected) fast retransmission	Sequence	TCP	40
Note	This frame is a (suspected) retransmission	Sequence	TCP	70
Note	Duplicate ACK (#1)	Sequence	TCP	846
Chat	TCP window update	Sequence	TCP	163
Chat	GET /openoffice/stable/2.0.0/OOo_2.0.0_Win32Intel_Install...	Sequence	HTTP	1
Chat	Connection establish acknowledge (SYN+ACK): server por...	Sequence	TCP	1
Chat	Connection establish request (SYN): server port 80	Sequence	TCP	1
Comment	Client: Window Scaling, SACK and a decent MSS... so far, s...	Comment	Frame	19

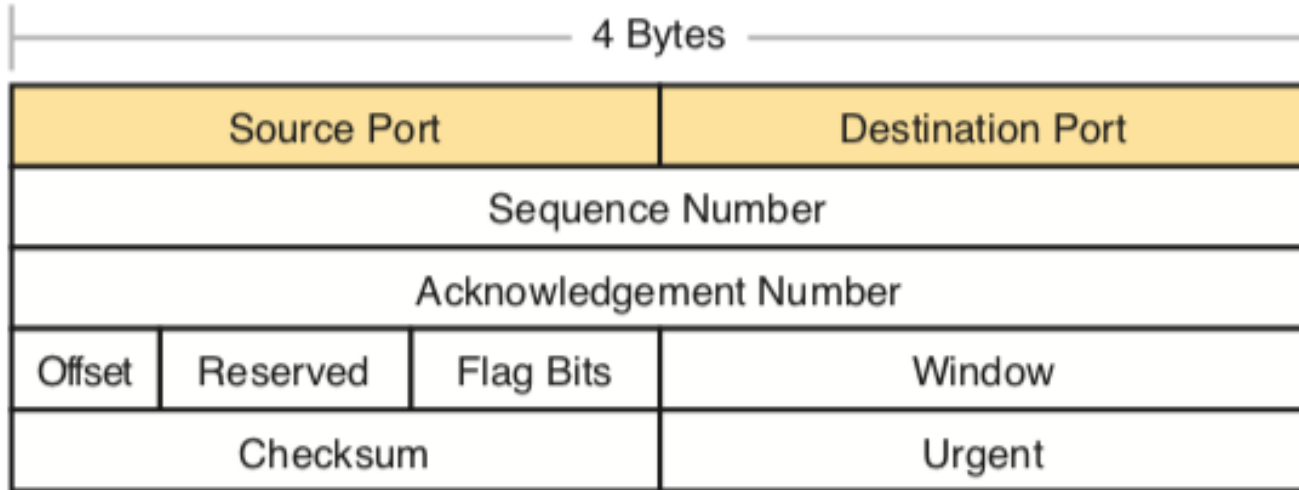
No display filter set.

☐ Limit to Display Filter ☒ Group by summary Search:

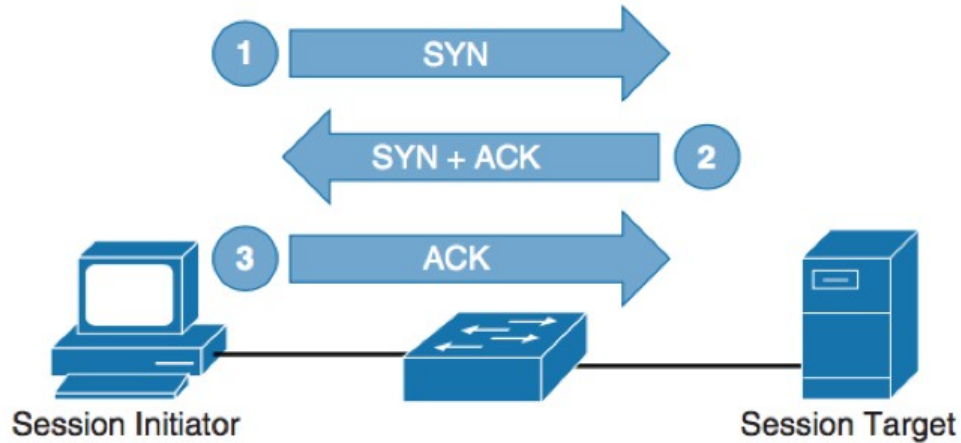
Review

TCP vs. UDP

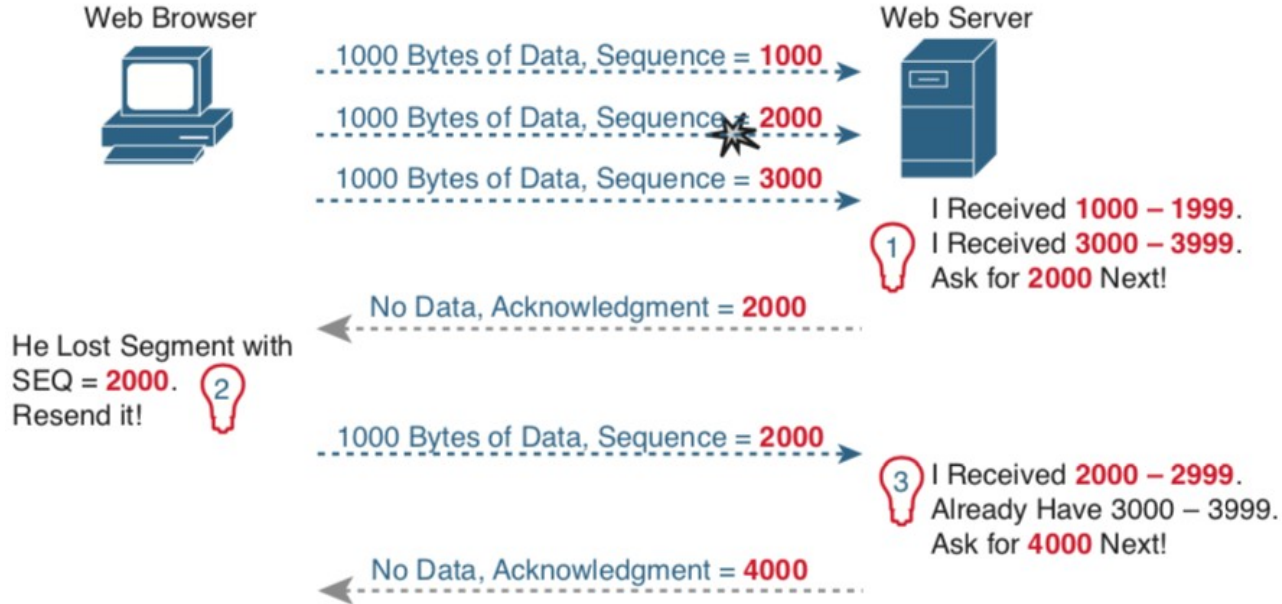
TCP Header Fields



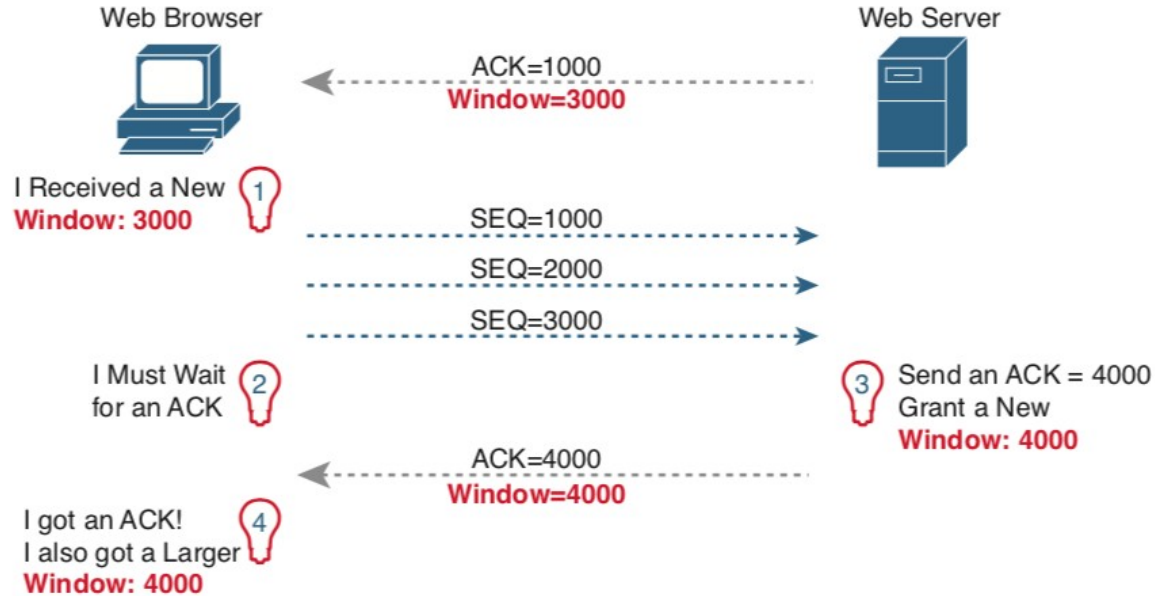
Connection-oriented__Three-way Handshake



TCP with Error Recovery



TCP Windowing



TCP vs UDP

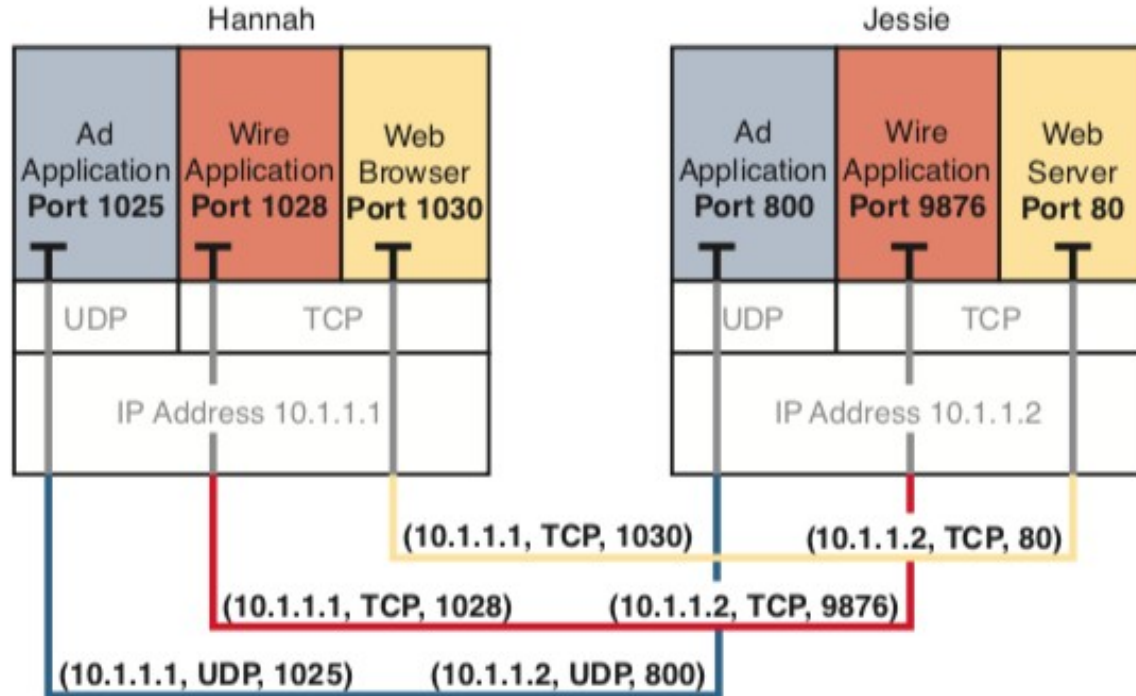
	TCP	UDP																								
Acronym	Transmission Control Protocol	User Datagram Protocol																								
Reliability	High	Best Effort																								
Header Size	4 Bytes <table><tr><td colspan="2">Source Port</td><td colspan="2">Destination Port</td></tr><tr><td colspan="4">Sequence Number</td></tr><tr><td colspan="4">Acknowledgement Number</td></tr><tr><td>Offset</td><td>Reserved</td><td>Flag Bits</td><td>Window</td></tr><tr><td colspan="2">Checksum</td><td colspan="2">Urgent</td></tr></table> Figure 5-1 TCP Header Fields 4x5=20 Byte	Source Port		Destination Port		Sequence Number				Acknowledgement Number				Offset	Reserved	Flag Bits	Window	Checksum		Urgent		4 Bytes <table><tr><td>Source Port</td><td>Destination Port</td></tr><tr><td>Length</td><td>Checksum</td></tr></table> Figure 5-10 UDP Header 4x2=8 Byte	Source Port	Destination Port	Length	Checksum
Source Port		Destination Port																								
Sequence Number																										
Acknowledgement Number																										
Offset	Reserved	Flag Bits	Window																							
Checksum		Urgent																								
Source Port	Destination Port																									
Length	Checksum																									

TCP vs UDP

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Sequence Number																														
Acknowledgement Number																														
Offset	Reserved	Flag Bits	Window																											
Checksum		Urgent																												
Source Port		Destination Port																												
Length		Checksum																												
Speed	Slow	Fast																												
Error Checking	Yes with Data Recovery	Yes But NO Data Recovery																												
Common Protocol	HTTP, HTTPS, FTP, Telnet	DNS, DHCP, TFTP, VOIP																												

Socket

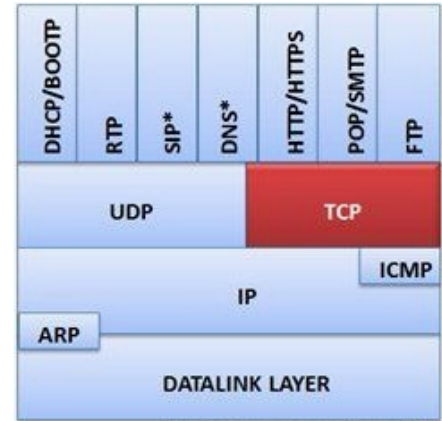
Connection between Sockets



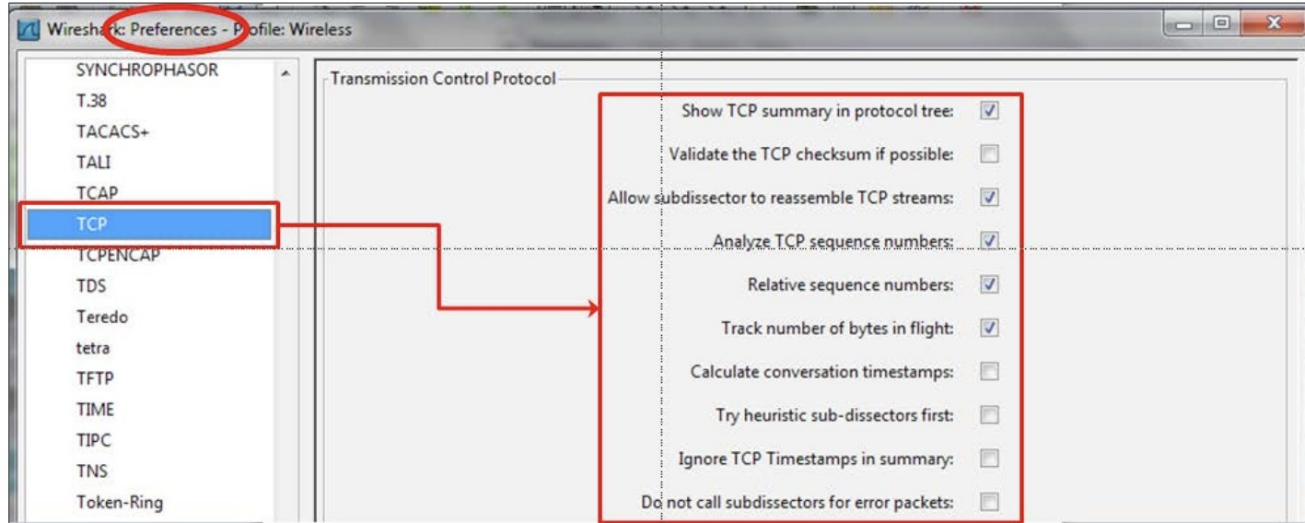
Analyze Transmission Control Protocol (TCP) Traffic

Transmission Control Protocol (TCP)

- **TCP offers a connection-oriented transport.**
- **Begins with a handshake between two devices.**
- **Data is sequenced and acknowledged**
 - To ensure proper delivery and automatic recovery for lost packets.
- **TCP supports windowing**
 - The process of sending numerous data packets in sequence without waiting for an intervening acknowledgment.



TCP Parameters

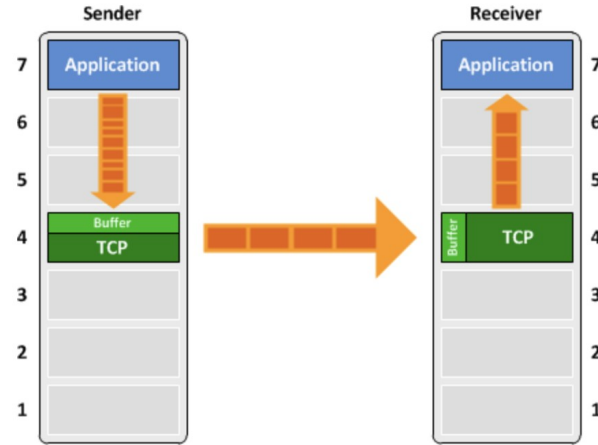


TCP Parameters

- ▶ **Show TCP summary in protocol tree:** Mark this button if you want the TCP summary line to be shown in the protocol tree (set by default).
- ▶ **Validate the TCP checksum if possible:** This feature can slow down performance. In most cases it is not required.
- ▶ **Allow subdissector to reassemble TCP streams:** This option is for stream analysis (set by default).
- ▶ **Analyze TCP sequence numbers:** When this is set, Wireshark analyzes sequence numbers and track phenomena such as retransmission, duplicate ACKs, and so on, which is one of the important features of Wireshark.
- ▶ **Relative sequence numbers:** When this is set, Wireshark will show you every TCP connection that starts from Seq=0.
- ▶ **Track number of bytes in flight:** This setting enables Wireshark to track the number of unacknowledged bytes flowing on the network (set by default).
- ▶ **Calculate conversation timestamps:** This feature enables the calculations of TCP timestamps option.
- ▶ **Try heuristic sub-dissectors first:** Try to decode a packet using heuristic method before using a sub-dissector registered to the specific port.
- ▶ **Ignore TCP Timestamps in summary:** Ignore the timestamp option in the TCP header.
- ▶ **Do not call subdissector for error packets:** This option does not analyze erroneous TCP packets.

The PSH Flag

Buffer generally has to get full being forwarded, but what if we didn't have any further data. That's how PSH flag comes handy..



TCP's push capability accomplishes two things:

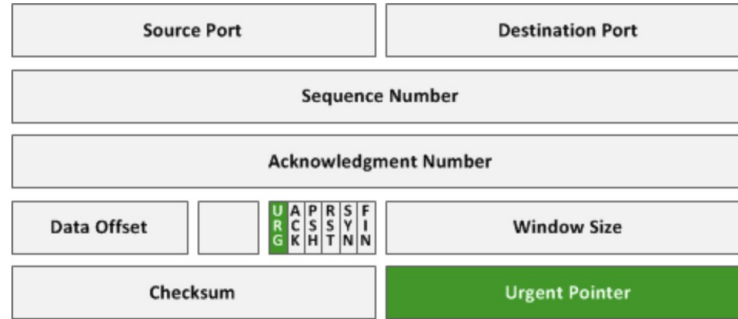
TCP buffers Data

Header length: 32 bytes
► Flags: 0x18 (PSH, ACK)
Window size: 6912 (scaled)

- The sending application informs TCP that data should be sent immediately.
- The PSH flag in the TCP header informs the receiving host that the data should be pushed up to the receiving application immediately.

The URG Flag

The URG flag is used to inform a receiving station that certain data within a segment is urgent and should be prioritized. If the URG flag is set, the receiving station evaluates the urgent pointer, a 16-bit field in the TCP header. This pointer indicates how much of the data in the segment, counting from the first byte, is urgent.



The URG flag isn't employed much by modern protocols

TCP Protocol Analysis

01

Observe

Sequence numbers count bytes; ACK numbers indicate the next expected byte.

02

Explain

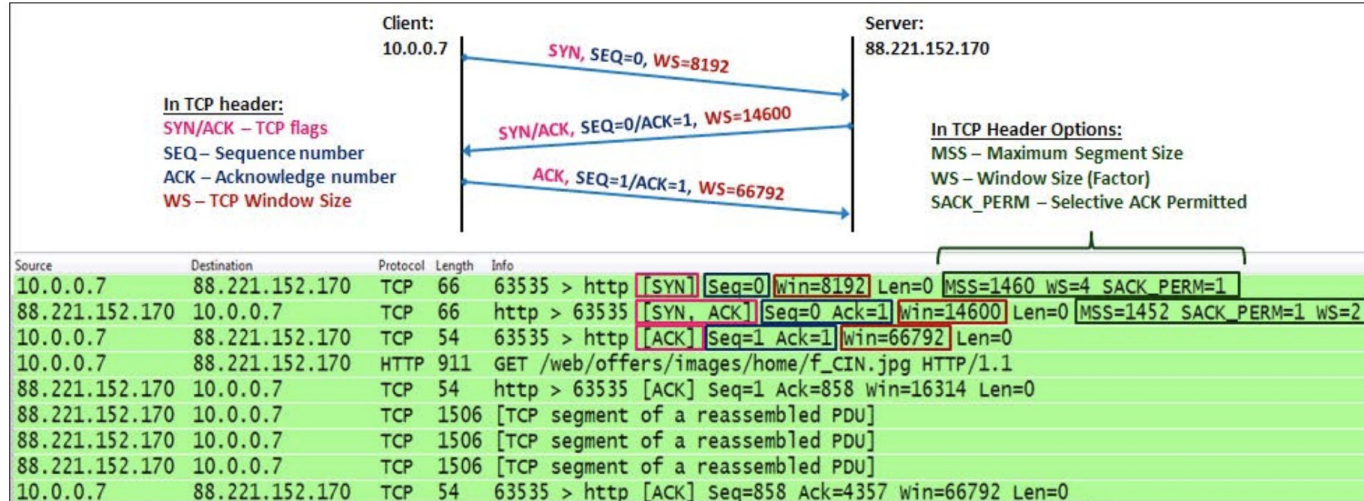
Retransmission and duplicate ACK labels are heuristics sensitive to capture placement and completeness.

03

Limit

Zero window indicates receive-side flow control; it is distinct from network congestion.

TCP Sequence and Acknowledgment



TCP Sequence and Acknowledgment

60	4.994751000	10.0.0.138	10.0.0.3	TCP	62 51910 > http [SYN] Seq=0 win=8192
61	5.088214000	10.0.0.3	81.218.31.171	TCP	62 51909 > http [SYN] Seq=0 win=8192
62	5.090244000	10.0.0.3	81.218.31.171	TCP	62 51912 > http [SYN] Seq=0 win=8192
63	5.178158000	10.0.0.3	81.218.31.171	TCP	66 51919 > xmpp-client [SYN] Seq=0 win=8192
64	6.247500000	10.0.0.3	173.194.78.125	TCP	66 51921 > http [SYN] Seq=0 win=8192
65	6.449442000	10.0.0.3	108.160.163.43	TCP	66 http > 51921 [SYN, ACK] Seq=0 Ack=54
66	6.480809000	108.160.163.43	10.0.0.3	TCP	54 51921 > http [ACK] Seq=1 Ack=1 Win=8192
67	6.480936000	10.0.0.3	108.160.163.43	TCP	54 http > 51921 [ACK] Seq=1 Ack=290 Win=8192
68	6.481512000	10.0.0.3	108.160.163.43	TCP	443 GET /subscribe?host_int=340855826&... HTTP/1.1 200 OK (text/html)
69	6.512241000	108.160.163.43	10.0.0.3	TCP	
70	6.512988000	108.160.163.43	10.0.0.3	TCP	

Connection not opened to 81.218.31.171 (SYN / SYN / SYN)

Connection opened to 108.160.163.43 SYN / SYN-ACK / ACK

No.	Time	Source	Destination	Protocol	Length	Info
2113	17.665372	10.0.0.3	135.82.12.1	TCP	66	62423 > http [SYN] Seq=0 win=8192
2120	17.746627	135.82.12.1	10.0.0.3	TCP	66	http > 62423 [SYN, ACK] Seq=0 Ack=54
2121	17.746693	10.0.0.3	135.82.12.1	TCP	54	62423 > http [ACK] Seq=1 Ack=1 Win=8192
2122	17.747085	10.0.0.3	135.82.12.1	HTTP	316	GET / HTTP/1.1
2130	17.862143	135.82.12.1	10.0.0.3	TCP	54	http > 62423 [ACK] Seq=1 Ack=290 Win=8192
2189	18.736301	135.82.12.1	10.0.0.3	TCP	145	[TCP segment of a reassembled stream]
2191	18.767301	135.82.12.1	10.0.0.3	TCP	1466	[TCP segment of a reassembled stream]

Filter: ip.addr==135.82.12.1

Connection opened to IP Address 135.82.12.1 TCP Port 80 (http)

RST/ACK Response

Here we see that when we try to connect to TCP port 6036, we get an RST/ACK response, which can be:

- ▶ A firewall that blocks port 6036 (that was the case here)
- ▶ When port address translation (PAT) is configured, and we translate only port 80 and not 6036
- ▶ The authentication of the username and password were done on TCP port 6036, the firewall allowed only port 80, the authentication was blocked, and the application didn't work

No.	Time	Source	Destination	Protocol	Length	Info
2620	36.423135	10.0.0.3	135.82.12.1	TCP	54	62438 > http [ACK] Seq=915
2621		10.0.0.3	135.82.12.1	TCP	66	62442 > 6036 [SYN] Seq=0 W
2622		135.82.12.1	10.0.0.3	TCP	54	6036 > 62442 [RST, ACK] Seq
2623		fe80::c067:2c23:335:ff02::c		SSDP	208	M-SEARCH * HTTP/1.1
2624		10.0.0.3	194.90.1.5	ICMP	74	Echo (ping) request id=0x0
2625		194.90.1.5	10.0.0.3	ICMP	74	Echo (ping) reply id=0x0
2626	37.329129	10.0.0.3	135.82.12.1	TCP	62	62442 > 6036 [SYN] Seq=0 W
2627	37.369547	135.82.12.1	10.0.0.3	TCP	54	6036 > 62442 [RST, ACK] Seq
2628	38.023274	10.0.0.3	194.90.1.5	ICMP	74	Echo (ping) request id=0x0

Connection Trials to TCP port 6036:

- SYN request
- RST/ACK response

Selective Acknowledgments

- Selective ACK is an option that enables the two parties of a connection to acknowledge specific packets, so when a single packet is lost, only this packet will be sent again.
- Must be set up during the TCP handshake process.

TCP Protocol Analysis

01

Observe

Sequence numbers count bytes; ACK numbers indicate the next expected byte.

02

Explain

Retransmission and duplicate ACK labels are heuristics sensitive to capture placement and completeness.

03

Limit

Zero window indicates receive-side flow control; it is distinct from network congestion.

TCP Recovery Mechanism

TCP Protocol Analysis

01

Observe

Sequence numbers count bytes; ACK numbers indicate the next expected byte.

02

Explain

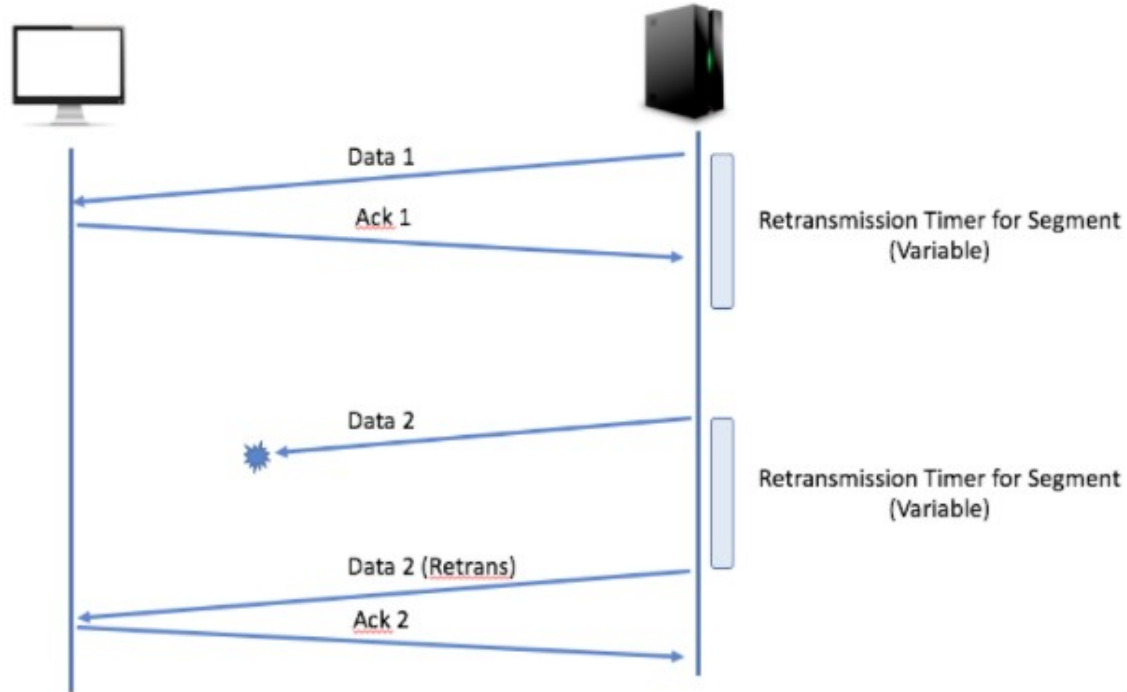
Retransmission and duplicate ACK labels are heuristics sensitive to capture placement and completeness.

03

Limit

Zero window indicates receive-side flow control; it is distinct from network congestion.

TCP Retransmissions



What Triggers TCP Retransmissions?

Reason	Solution
????	
????	
????	

What Triggers TCP Retransmissions?

Reason	Solution
packet loss	Since packet loss typically occurs at interconnecting devices, you should move your analyzer around a bit to try to isolate the offending device.
retransmission timeout (RTO) timer expires (next page)	
receiver sends Duplicate Acknowledgments to request a missing segment	
Filter → tcp.analysis.retransmission	

TCP Protocol Analysis

01

Observe

Sequence numbers count bytes; ACK numbers indicate the next expected byte.

02

Explain

Retransmission and duplicate ACK labels are heuristics sensitive to capture placement and completeness.

03

Limit

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TCP Protocol Analysis

01

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02

Explain

Retransmission and duplicate ACK labels are heuristics sensitive to capture placement and completeness.

03

Limit

Zero window indicates receive-side flow control; it is distinct from network congestion.

TCP Fast Retransmit and Recovery (FRR)

- FRR is a congestion control algorithm that makes it possible to quickly recover lost data packets.
- Without FRR, the TCP uses a timer that requires a RTO if a packet is lost. No new or duplicate packets can be sent during the timeout period.
- With FRR, if a receiver receives a data segment that is out of order, it immediately sends a duplicate acknowledgement to the sender. If the sender receives three duplicate acknowledgements, it assumes that the data segment indicated by the acknowledgements is lost and immediately retransmits the lost segment.
- With FRR, time is not lost waiting for a timeout in order for retransmission to begin.

What Triggers ACKed Lost Packet?

Reasons

????

????

What Triggers ACKed Lost Packet?

Reasons

(1) network supports multiple paths

(asymmetrical routing)

(2) capture process is faulty

e.g. Overloaded span switch port may not forward all the traffic down the monitor port.

What Triggers Duplicate ACK?

If a packet is detected as missing (the expected sequence number is skipped), the receiver generates an ACK.

indication:

A high number of Duplicate ACKs may be an indication of **high latency** between TCP hosts as well as packet loss.

SEE file → [http-download-bad.pcapng](#)

What Triggers Zero Window?

Reason	Try it out!
When a receiver has no receive buffer space available! WHY?	File → http-download-bad.pcapng



What Triggers Zero Window?

Reason	Try it out!
When a receiver has no receive buffer space available! WHY?	File → http-download-bad.pcapng



TCP Protocol Analysis

01

Observe

Sequence numbers count bytes; ACK numbers indicate the next expected byte.

02

Explain

Retransmission and duplicate ACK labels are heuristics sensitive to capture placement and completeness.

03

Limit

Zero window indicates receive-side flow control; it is distinct from network congestion.

What Triggers **Out-of-Order**?

1. Multiple Paths

data traffic travels along multiple paths to get to the destination. Data streams traveling along multiple paths may encounter different latency times, thus triggering unnecessary retransmissions if a receiver times out waiting for a packet.

2. Queuing Device

queuing device along a path does not forward packets in the same order in which they arrived. For example, packets arrive at the queuing device in 1-2-3-4 order, but are queued and forwarded in 1-3-2-4 order.

Note → An application may not be affected by this slight reordering, but it is not an ideal way for a queuing device to behave.

TCP Protocol Analysis

Observe

Sequence numbers count bytes; ACK numbers indicate the next expected byte.

Interpret

Retransmission and duplicate ACK labels are heuristics sensitive to capture placement and completeness.

Corroborate

Zero window indicates receive-side flow control; it is distinct from network congestion.

Lab 14 • Investigate retransmission and zero window

01

Build

A reproducible packet table and written findings.

02

Evidence

1 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

15 Traffic Graphs

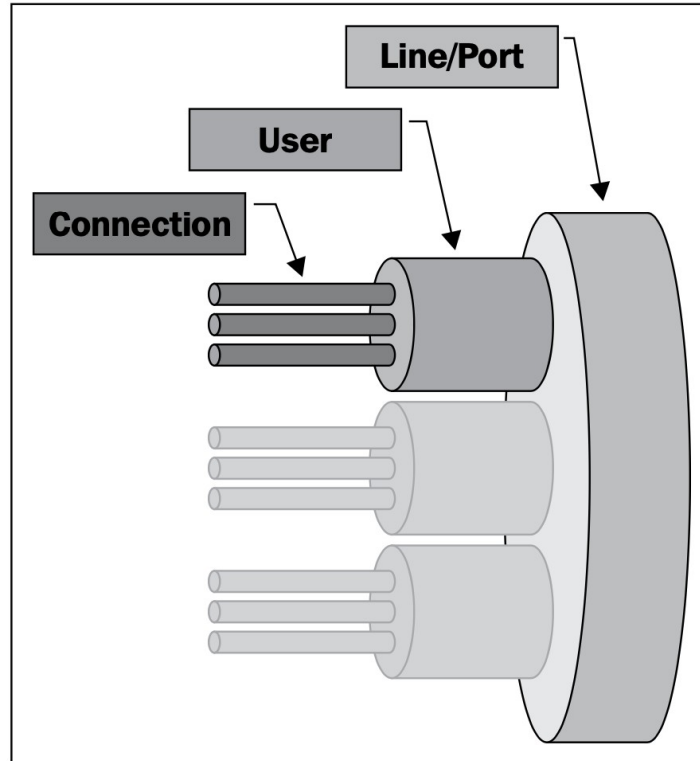
An I/O graph bins observed events; units and interval determine what it means.

Lab 15 **Graph bytes, RTT and recovery**

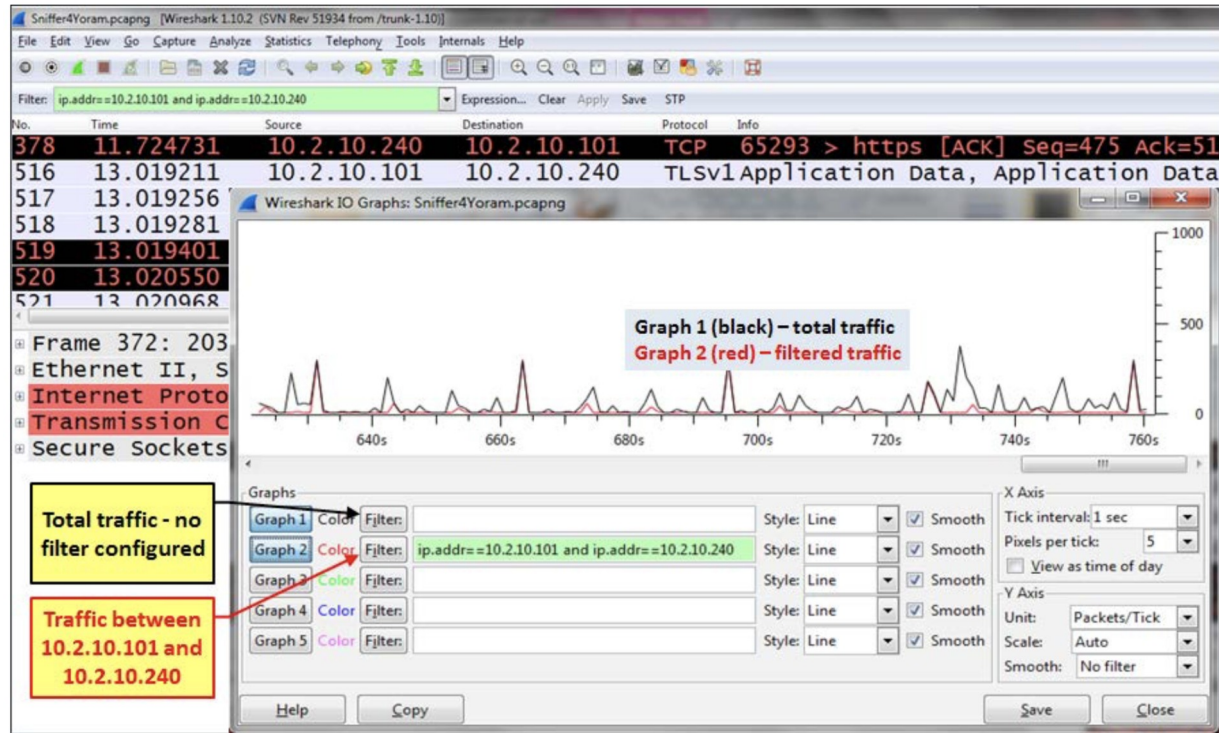
Use the self-contained lab folder and Learner Guide. Observe → explain → verify → discuss.

Graph I/O Rates and TCP Trends

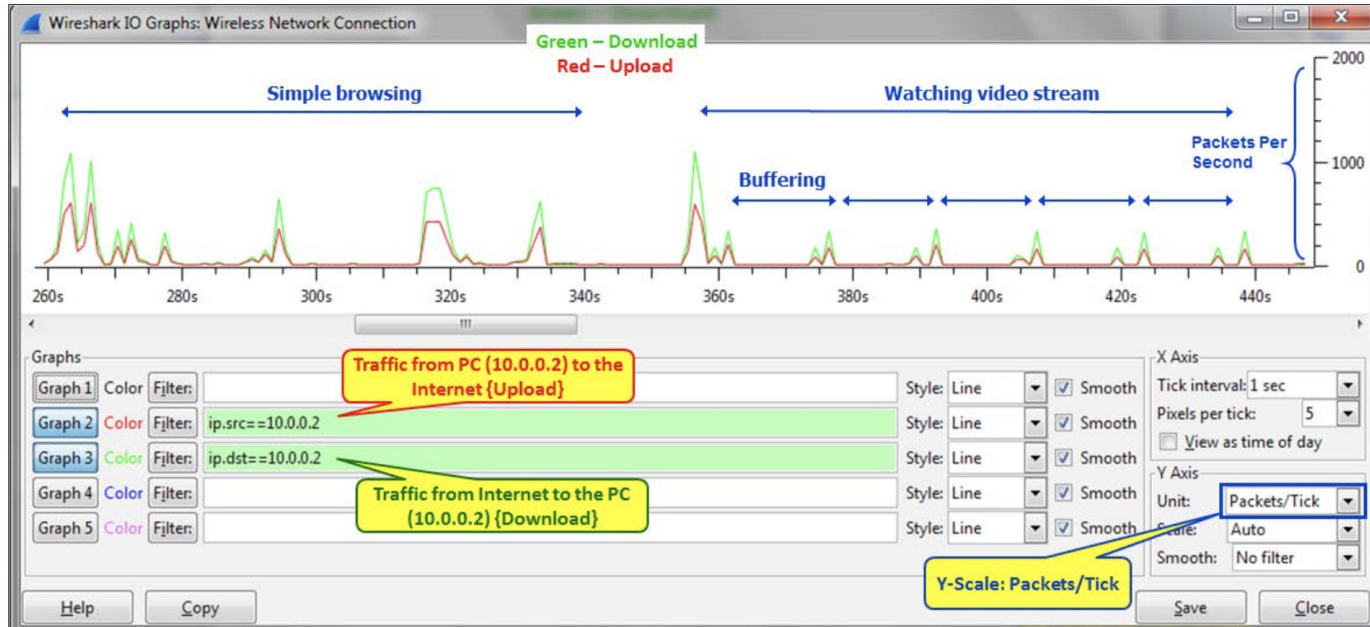
Measuring Throughput Between End Devices



IO Graph__Example #1



IO Graph__Example #2

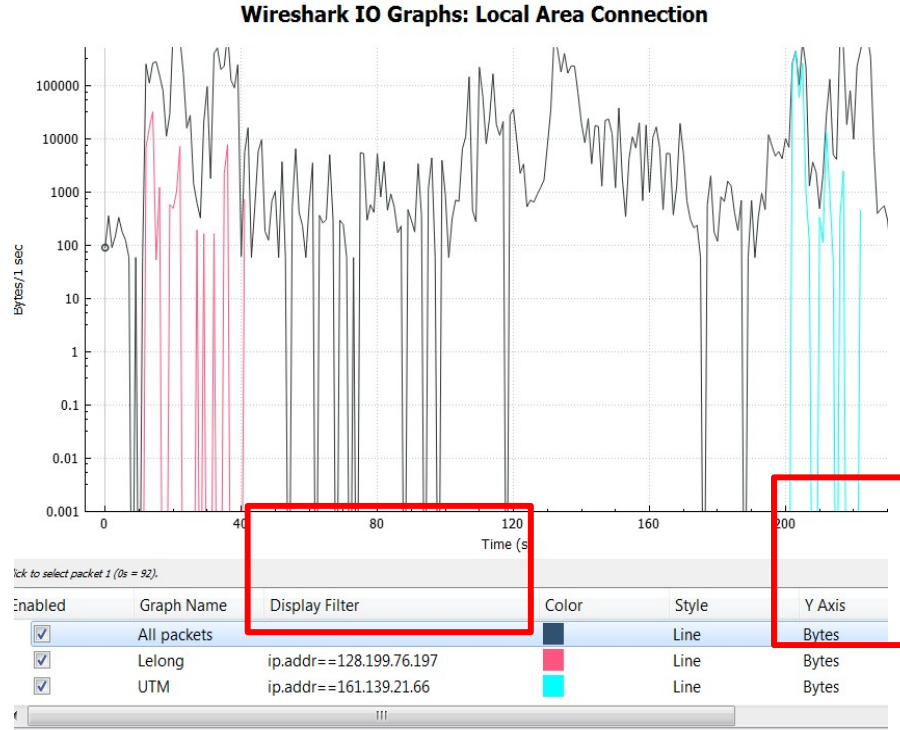


IO Graph__Example #3



In the second graph, we see the traffic in bits/sec. Here, we see the bandwidth required from the network while using it to connect to the Internet; that is, an asymmetrical bandwidth when most of the traffic is in the download direction.

Advanced IO Graphs



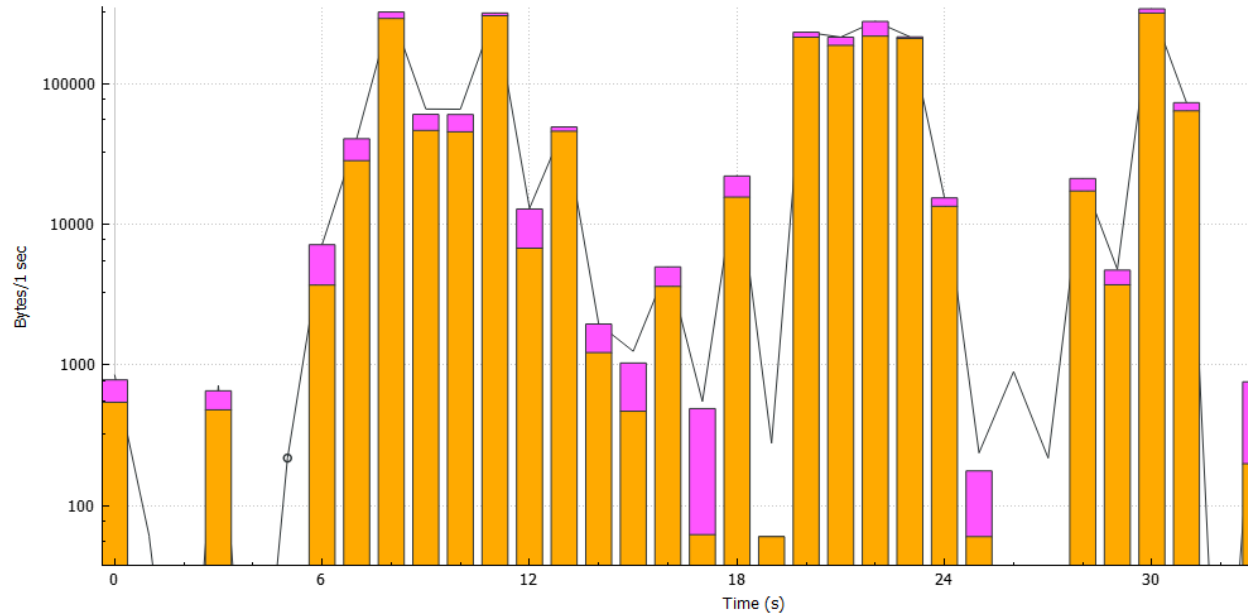
Note:

Use domain name instead

http.host matches "www.yahoo.com"

However this only show TO this domain traffics.

Wireshark IO Graphs: Local Area Connection

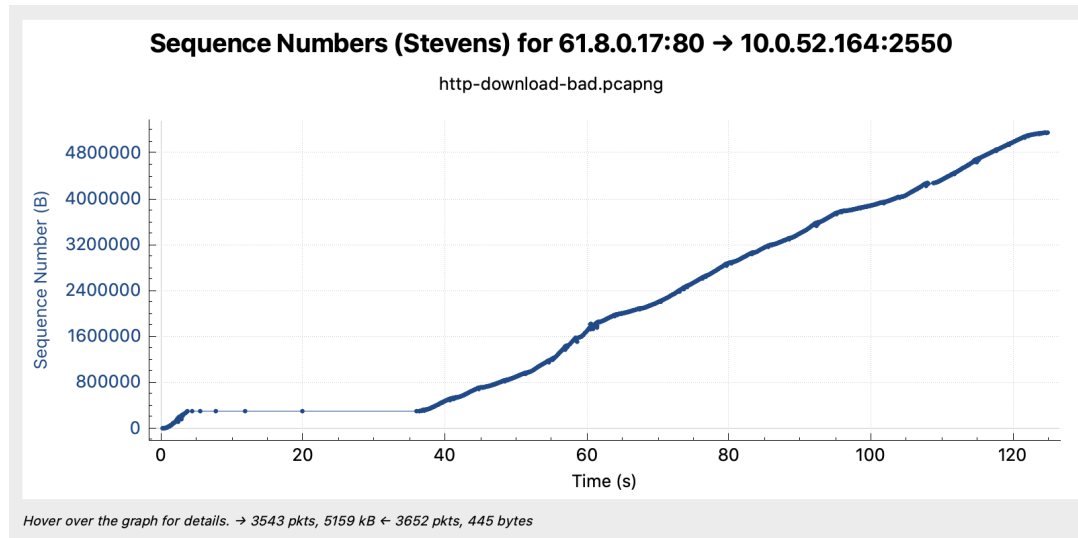


Click to select packet 11 (5s = 217).

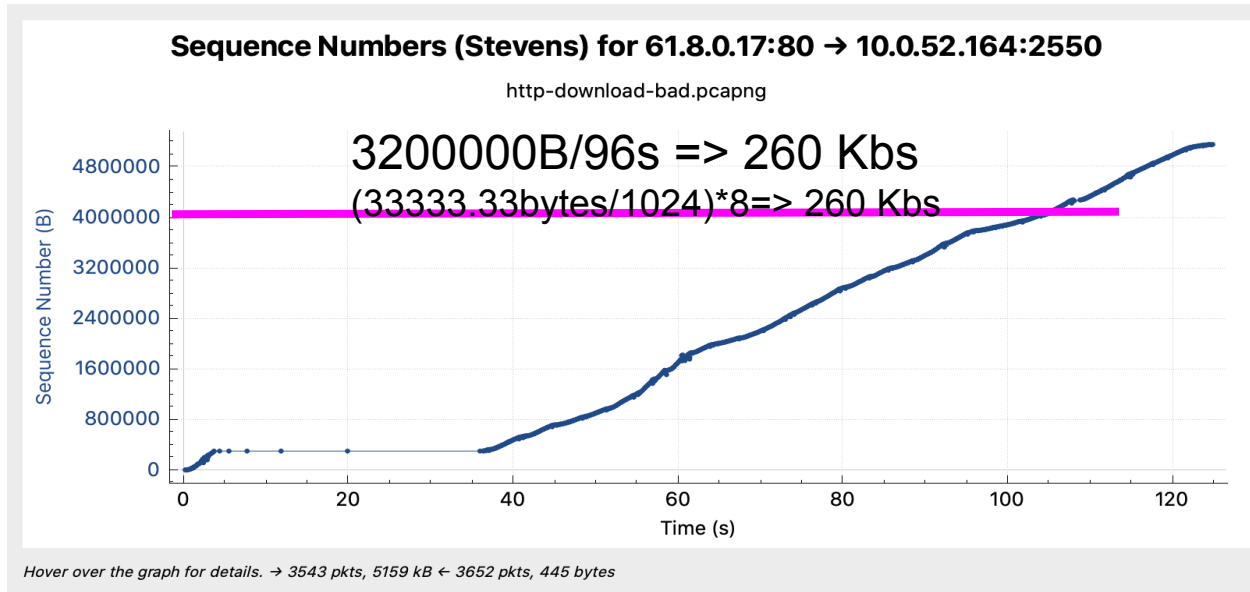
Enabled	Graph Name	Display Filter	Color	Style	Y Axis	Y Field
☒	All packets			Line	Bytes	
☒	Sent	ip.src == 10.2.1.35		Stacked Bar	Bytes	
☒	Received	ip.dst == 10.2.1.35		Stacked Bar	Bytes	

TCP Stream Graph

How to Measure the Throughput of A File Transfer by Using Time/Sequence (Stevens) Graph?



How to Measure the Throughput of A File Transfer by Using Time/Sequence (Stevens) Graph?



Traffic Graphs

Observe

An I/O graph bins observed events; units and interval determine what it means.

Interpret

TCP time-sequence graphs show byte progress, stalls and repeated sequence ranges.

Corroborate

RTT samples, application response time and throughput answer different questions.

Lab 15 • Graph bytes, RTT and recovery

01

Build

A reproducible packet table and written findings.

02

Evidence

2 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

16 HTTP and HTTP/2 Analysis

HTTP response codes describe application outcomes after transport delivery.

Lab 16 **Inspect HTTP outcomes and HTTP/2 frames**

Use the self-contained lab folder and Learner Guide. Observe → explain → verify → discuss.

Follow Streams and Reassemble Data

HTTP and HTTP/2 Analysis

Observe

HTTP response codes describe application outcomes after transport delivery.

Interpret

A response time contains more than network latency; pair the correct request and response.

Corroborate

HTTP/2 multiplexes frames; encrypted HTTP/2 needs authorised TLS secrets to inspect payload.

Lab 16 • Inspect HTTP outcomes and HTTP/2 frames

01

Build

A reproducible packet table and written findings.

02

Evidence

2 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

17 TLS-Encrypted Traffic Analysis

Without session secrets, encrypted application records do not disclose HTTP payload.



Lab 17 • Compare encrypted and decrypted views

TLS-Encrypted Traffic Analysis

Observe

Without session secrets, encrypted application records do not disclose HTTP payload.

Interpret

A controlled key log enables authorised inspection of this synthetic TLS exchange.

Corroborate

A server RSA private key alone cannot decrypt modern ECDHE or TLS 1.3 sessions.

Lab 17 • Compare encrypted and decrypted views

01

Build

A reproducible packet table and written findings.

02

Evidence

1 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

18 Ten Troubleshooting Steps and Reporting

Begin with scope and a baseline, then focus with endpoints, filters, time and graphs.

Lab 18 **Produce an evidence-led incident report**

Use the self-contained lab folder and Learner Guide. Observe → explain → verify → discuss.

Annotate, Save, Export and Print Packets

Ten Troubleshooting Steps and Reporting

Observe

Begin with scope and a baseline, then focus with endpoints, filters, time and graphs.

Interpret

Separate observed facts from hypotheses and specify what new evidence would test them.

Corroborate

A useful report states affected flow, measurement, impact, recommended action and uncertainty.

Lab 18 • Produce an evidence-led incident report

01

Build

A reproducible packet table and written findings.

02

Evidence

1 frame(s) match the core question in the supplied synthetic fixture.

03

Reflect

Which second observation would strengthen your conclusion?

What You Achieved

01

Capture

Planned scope and a repeatable analyst workflow.

02

Diagnose

Interpreted protocol, timing and application evidence.

03

Report

Separated facts, hypotheses and next actions.

Keep Practising

Repeat

Redo a lab with a fresh profile.

Apply

Use the checklist with an authorised new capture.

Review

Check that every conclusion cites a packet or measurement.

Thank You

**Keep the evidence.
Explain the uncertainty.**